

LegalEase.AI

Product Design & Technical Architecture Documentation Suite

PRD | System Architecture | Database Design | API Reference

AI Architecture | Security | Deployment | Workflows

Competitive Analysis | Technical Deep Dive | Roadmap

Document Version 1.0 | Generated 05 June 2026 (UTC)

Classification: Confidential - Internal & Investor Use

Table of Contents

1. Executive Summary
2. Document 1 - Product Requirements Document (PRD)
3. Document 2 - System Architecture
4. Document 3 - Database Design
5. Document 4 - API Documentation
6. Document 5 - AI Architecture
7. Document 6 - Security Architecture
8. Document 7 - Complete Deployment Guide (Laptop + EC2 + Docker)
9. Document 8 - Product Workflow Guide
10. Document 9 - Competitive Analysis
11. Document 10 - Technical Deep Dive
12. Appendix A-L - Glossary, Roadmap, Related Docs
13. Appendix D - Expanded Module Specifications
14. Appendix E-U - API Index, DB, UI, Runbooks, FAQ
15. Appendix V - Complete API Route Index (493 routes, auto-generated)
16. Appendix W - Complete Environment Variables Catalog

Table of Contents

- 1. Executive Summary
- 2. Document 1 - Product Requirements Document (PRD)
- 3. Document 2 - System Architecture
- 4. Document 3 - Database Design
- 5. Document 4 - API Documentation
- 6. Document 5 - AI Architecture
- 7. Document 6 - Security Architecture
- 8. Document 7 - Deployment Architecture
- 9. Document 8 - Product Workflow Guide
- 10. Document 9 - Competitive Analysis
- 11. Document 10 - Technical Deep Dive
- 12. Product Roadmap & Appendix

Executive Summary

LegalEase.AI is an AI-powered legal intelligence platform purpose-built for Indian law practice. It unifies document-grounded research (RAG), live statutory and case-law research (Open Law / Hybrid), matter-centric case management, drafting studio, CRM intake, billing, litigation operations, enterprise DMS, and an Evidence Intelligence Center (e-discovery) into a single SaaS product.

This documentation suite describes the complete product vision, technical architecture, data model, API surface (503+ HTTP routes), AI pipelines, security posture, deployment topology, user workflows, competitive positioning, and engineering rationale. It is intended for investors, enterprise buyers, technical interviewers, and engineering teams onboarding to the codebase.

****Production URL:****

<https://legalease.duckdns.org>

****Stack:**** Next.js 15 + FastAPI +

PostgreSQL/SQLite + FAISS + Redis + Docker
+ nginx

****AI:**** Ollama (local KB) + Gemini (web research) + adaptive learning feedback loops

Document 1 - Product Requirements Document (PRD)

1.1 Vision & Problem Statement

Indian legal practice is fragmented across disconnected tools: generic chatbots without citations, DMS without AI, research platforms without matter context, and practice management without intelligence. Lawyers lose hours re-reading PDFs, manually mapping IPC to BNS, triaging evidence, and drafting repetitive pleadings.

LegalEase.AI solves this by anchoring every AI answer to ****evidence**** (uploaded documents, matter files, firm knowledge base) while providing ****live research**** when internal documents are insufficient. The platform learns from attorney feedback without requiring GPU fine-tuning on every interaction.

1.2 Target Users

Persona | Needs | Primary Modules

Solo / Small Firm Lawyer | KB chat, drafting, IPC-BNS, billing | AI Assistant, Drafting, Legal Tools, Billing

Litigation Associate | Hearings, cause lists, evidence, timelines | Matters, Litigation Desk, Evidence Intelligence

Firm Partner / Owner | Team, billing, analytics, enterprise DMS | Dashboard, Enterprise, Admin, Analytics

Paralegal / Intern | Intake, triage, document upload | Intake Desk, Documents, Discovery

Client (portal) | Status, uploads, approvals | Client Portal (token-based)

1.3 Product Modules (Feature Inventory)

Core Platform

****AI Assistant**** - Multi-mode chat:
Knowledge Base, Open Law (web), Hybrid,
Deep Case

****Documents & KB**** - PDF/image upload,
OCR, chunking, FAISS indexing, reindex
jobs

****Matters**** - Case workspace: documents,
timeline, hearings, tasks, evidence,
entities, contradictions

****Memory**** - Persona, facts, thread
summaries, past-conversation RAG

****Adaptive Learning**** - Thumbs feedback,
chunk boosts, query expansion, coach
pipeline

Practice Operations

****Intake Desk (CRM)**** - Lead pipeline,
Kanban, AI classification, matter
conversion

****Billing**** - Time entries, expenses,
invoices, PDF export, Stripe
subscriptions

****Trust Ledger**** - Client trust accounts
and transactions

****Litigation Desk**** - Cause lists, court
day, eCourts sync, war room, limitation
calculator

Content & Drafting

****Drafting Studio v2/v3/v4**** - Templates,
versions, redline, copilot, court
bundles, signatures

****Clause Library**** - Reusable clauses
with feedback loop

****Legal Tools**** - IPC to BNS converter
(official dataset), limitation presets

Enterprise

****Enterprise Workspace**** - DMS folders,

court orders, knowledge entries, client portal ops

****Firm Chat**** - Real-time collaboration, matter-linked rooms, SSE/WebSocket

****Evidence Intelligence Center**** - Upload PDF/DOCX/email/ZIP, OCR, classification, entity extraction, timeline, privilege detection, statute finder, contradiction detector

Premium / Advanced

Witness Simulator, Deal Rooms, BNS Auditor, PII Redactor, Judicial Analytics

1.4 User Journeys

Journey A - New matter to grounded answer

1. Create matter (client, practice area, court)
2. Upload case PDFs to matter or global KB
3. Wait for indexing (ML worker / inline job)
4. Open AI Assistant, select matter scope
5. Ask legal question in Knowledge Base mode
6. Receive cited answer with source filename and section
7. Thumbs up/down trains retrieval for future queries

Journey B - Evidence investigation

1. Open Evidence Intelligence Center (/discovery)
2. Link matter from dropdown
3. Drag-and-drop vendor emails, invoices, WhatsApp exports (ZIP)
4. System runs OCR, extracts entities and dates
5. Review Evidence Strength score, privilege flags, risk indicators
6. Inspect auto-built timeline across all

matter evidence

7. Run contradiction check between two witness statements

8. Export findings to matter repository

Journey C - Client intake to matter

1. Lead arrives via public intake form or manual entry

2. CRM AI classifies practice area and urgency

3. Paralegal moves lead through Kanban stages

4. Convert lead to matter with one click

5. Assign lawyer, seed folders, link documents

Journey D - Drafting to filing

1. Create draft from template in Drafting Studio

2. AI autofill from matter variables

3. Clause intelligence and risk scan

4. Partner review workflow, track changes

5. Export PDF/DOCX court bundle

6. Optional e-sign request to client portal

1.5 Success Metrics (KPIs)

Metric | Target | Measurement

KB answer citation rate | >85% substantive answers include source | learning analytics

NOT_FOUND precision | Low false negatives on indexed docs | smoke tests

Time to first indexed doc | <3 min for 50-page PDF | index job telemetry

Feedback adoption | >30% sessions with thumbs | adaptive_interactions

Matter conversion (CRM) | >25% qualified leads | crm_leads pipeline

Uptime (production) | 99.5% | health/live probes

Gemini daily quota compliance | Zero overage on Free tier | gemini_usage_daily

1.6 Competitors (Summary)

See Document 9 for full comparison.

LegalEase differentiates on Indian statute tooling (IPC-BNS), integrated matter+AI, Evidence Intelligence, and affordable SaaS pricing vs Harvey AI or Relativity.

1.7 Out of Scope (Current Phase)

Native iOS/Android apps (responsive web first)

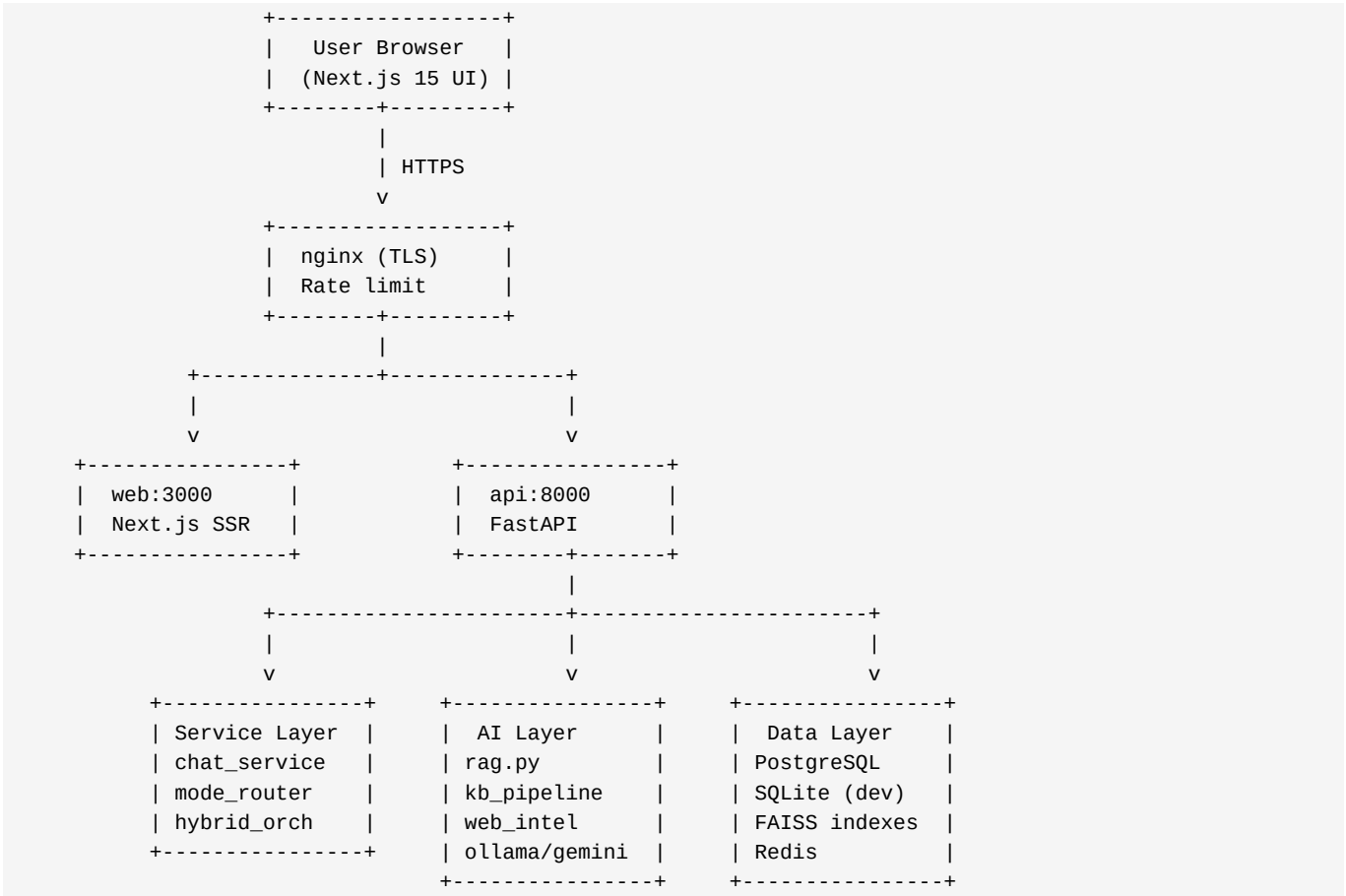
Live eCourts API write-back (read/sync only)

Full Relativity-scale distributed processing cluster

E2E encrypted chat incompatible with server-side RAG

Document 2 - System Architecture

2.1 High-Level Architecture



2.2 Layer Responsibilities

Layer	Location	Responsibility
Frontend	web/app/(app)/	Routes, chat UI, matter workspace, settings
API Gateway	backend/app/main.py	Auth, middleware, CORS, health
Routers	backend/app/api/v1/endpoints/	38 endpoint modules, 503 routes
Services	backend/app/services/	Chat orchestration, mode routing, open law
Core	backend/app/core/	Domain logic, repos, schemas, learning
Legacy	legacy_saas/, rag.py, kb_pipeline.py	Auth, RAG engine, PDF extraction

2.3 Product Architecture Diagram

```
Users
|
v
Frontend (Next.js)
|
v
Backend (FastAPI)
|
+-- AI Layer (RAG, Open Law, Hybrid, Coach)
+-- Legal Tools (IPC-BNS, Limitation, Court Fees)
+-- Drafting Studio (v2/v3/v4 lifecycle)
+-- Enterprise (DMS, Court Orders, Knowledge)
+-- Litigation (Cause lists, War room, eCourts)
+-- Evidence Intelligence (OCR, classification, timeline)
+-- Analytics & Learning (feedback, tuning export)
```

2.4 AI Assistant Architecture

```
User Query
|
v
Intent Detection (intent_engine.py, query_parser)
|
v
Mode Router (mode_router.py)
|
+-- knowledge_base --> kb_pipeline --> rag.py --> FAISS --> Ollama LLM
|
+-- open_law/web --> web_provider_chain --> Gemini grounded search
|                                     --> Tavily/Serp fallback
|                                     --> Ollama fallback
|
+-- hybrid/deep_case --> hybrid_orchestrator --> KB chunks + web fusion
|
v
Response + citations + follow-ups + interaction_id (learning)
```

****Key policy:**** Gemini is blocked from synthesizing KB answers (kb_gemini_safety.py). KB mode uses Ollama + indexed PDFs only.

2.5 Drafting Studio Architecture

```
Template / Smart Draft
|
v
Clause Engine (clause_library, drafting_v3 clause-intel)
|
v
AI Generate / Copilot (drafting_workspace, llm_orchestrator)
|
v
Risk Scanner + Document Health (drafting_v3 insights, filing-readiness)
```

```
|
v
Version Control (workspace_draft_versions)
|
v
Review Workflow (partner-review, track-changes, signatures)
|
v
Export (PDF/DOCX court bundle, drafting_docx_export)
```

2.6 Legal Tools (IPC-BNS) Architecture

```
User Query / Document Upload
|
v
IPC-BNS Engine v3 (ipc_bns_engine_v3.py)
|
v
Official Verified Dataset (BNS mappings)
|
v
Search + Compare + Bulk Convert
|
v
Mapping Validation + Report Export (PDF/DOCX)
```

2.7 Evidence Intelligence Architecture

```
Evidence Upload (PDF/DOCX/EML/ZIP/Image)
|
v
OCR + Text Extraction (evidence_extraction.py, pdf_extraction, ocr_engine)
|
v
Metadata Extraction (author, dates, SHA-256 hash)
|
v
AI Classification (FINANCIAL, CONTRACT, COMMUNICATION, COURT_ORDER, ...)
|
v
Entity Extraction (people, orgs, dates, phones, IFSC, case numbers)
|
v
Risk + Privilege Detection
|
v
Timeline Builder + Statute Finder (BNS/BNSS)
|
v
PostgreSQL discovery_items + Matter Linkage
```

2.8 Enterprise Workflow

```
Client Portal (token URL)
  |
  v
Matter
  |
  v
DMS (ent_dms_folders, ent_dms_documents)
  |
  v
Court Orders (ent_court_orders + AI analysis)
  |
  v
Knowledge Base (ent_knowledge)
  |
  v
Analytics (dashboard, saas-metrics)
```

2.9 Background Workers

Worker | Script | Queue

E-Discovery | scripts/ediscovery_worker.py | Redis
legalease:ediscovery:queue

ML Indexing | scripts/ml_worker.py | Redis ML queue

Coach scheduler | backend/app/core/coach_scheduler.py
| Cron-style

Document 3 - Database Design

3.1 Storage Strategy

Environment | Primary DB | Vector Store | Sessions

Laptop dev | SQLite (legalease.db) | FAISS local dirs
| In-memory / SQLite

Production EC2 | PostgreSQL 16 | FAISS on Docker
volume | Redis 7

****Flag:**** SAAS_USE_POSTGRES_LEGACY=1
routes all legacy tables to PostgreSQL via
connect_app_db().

3.2 Entity Relationship Overview

```
User (users)
|
+-- Organization (organizations) -- org_members
|
+-- Matter (matters)
|   |
|   +-- Documents (documents, ent_dms_documents)
|   +-- Hearings (matter_hearings)
|   +-- Tasks (matter_tasks)
|   +-- Evidence (matter_evidence, discovery_items)
|   +-- Timeline (matter_timeline)
|   +-- Entities (matter_entities)
|   +-- Contradictions (matter_contradictions)
|
+-- Chat (chat_history, adaptive_interactions)
+-- Drafts (workspace_drafts, workspace_draft_versions)
+-- CRM (crm_leads + v2 extensions)
+-- Billing (financial_records, invoices, trust_accounts)
```

3.3 Core Tables (pg_core_schema)

Table | Purpose | Key Fields

users | Authentication | id, username, password_hash,
membership, role

organizations | Multi-tenant firms | org_id, name,
plan, seat_limit

org_members | Team roster | org_id, user_id, role

matters | Case header | matter_id, user_id,
matter_name, practice_area

chat_history | Persisted turns | id, user_id,
question, answer, thread_id, matter_id

adaptive_interactions | Learning log | id, user_id,
mode, query, chunk_keys

adaptive_feedback | Thumbs/events | id,
interaction_id, signal, value
kb_answer_memory | Cached KB successes | query_norm,
answer, confidence

3.4 Practice / Matter Tables (practice_schema)

Table | Purpose

matter_notes | Internal case notes
matter_timeline | Chronological events
matter_hearings | Court dates, cause list imports
matter_tasks | Task management
matter_deadlines | Limitation and filing deadlines
matter_entities | Parties, witnesses
matter_evidence | Matter-scoped evidence records
matter_contradictions | Detected conflicts
matter_members | RBAC on matter
matter_audit_log | Immutable audit trail
workspace_drafts | Drafting documents
workspace_draft_versions | Version history

3.5 SaaS / Operations Tables (saas_schema)

Table | Purpose

financial_records | Time billing entries
invoices | Invoice lifecycle
crm_leads | Intake pipeline
ediscovery_batches | Evidence batch uploads
discovery_items | Analyzed evidence items (+
metadata_json, entities_json, timeline_json)
discovery_tag_weights | Learned tag weights per
matter
research_queries | Research session logs
trust_accounts / trust_transactions | Client trust
ledger
client_portal_access | Magic-link tokens

3.6 Enterprise Tables (ent_*)

Table | Purpose

ent_dms_folders | Folder hierarchy
ent_dms_documents | Document metadata + OCR text
ent_dms_versions | Version control
ent_court_orders | Ingested orders + AI summary
ent_knowledge | Firm knowledge entries
ent_client_requests | Portal document requests

3.7 Vector Index Layout

```
faiss_indexes/  
  user_{user_id}/  
    global_kb/      # Statutes, firm-wide uploads  
    matter_{matter_id}/ # Matter-scoped evidence only
```

Matter AI mode queries matter index exclusively (kb_retrieval_router.py).

3.8 Indexing & Constraints

Foreign keys on SQLite; UNIQUE constraints on Postgres org_members, discovery_tag_weights

Soft delete on matters (is_archived flag)

CASCADE delete on discovery_items when batch removed

JWT membership refreshed from DB on each request (not trusted from token alone for plan gating)

Document 4 - API Documentation

4.1 Base URL & Authentication

Environment | Base URL

Production | <https://legalease.duckdns.org/api/v1>

Local dev | <http://127.0.0.1:8000/api/v1>

****Authentication:**** Bearer JWT from POST
`/api/v1/auth/login`

POST `/api/v1/auth/login`

Content-Type: `application/json`

```
{"username": "lawyer@firm.com", "password": "..."}

```

Response 200:

```
{"token": "<jwt>", "user_id": "...", "membership": "Pro"}

```

GET `/api/v1/auth/me`

Authorization: Bearer `<jwt>`

****Error codes:**** 401 Unauthorized, 403 Forbidden (plan/suspension), 404 Not found, 422 Validation, 429 Rate limit, 500 Server error

4.2 Chat API

POST `/api/v1/chat`

Authorization: Bearer `<token>`

Content-Type: `application/json`

```
{
  "message": "Explain IPC Section 300",
  "mode": "knowledge_base",
  "lang": "English",
  "history": [],
  "matter_id": "optional-uuid",
  "thread_id": "optional-uuid"
}

```

****Response fields:**** content,
similar_cases, web_sources, follow_ups,
session_id, chat_id, thread_id,
interaction_id

POST `/api/v1/chat/stream`

Server-Sent Events stream for token-by-token rendering.

4.3 Matters API (Selected)

Method | Path | Purpose

GET | /matters | List matters
POST | /matters | Create matter
GET | /matters/{id} | Matter detail
PATCH | /matters/{id} | Update matter
GET | /matters/{id}/timeline | Timeline events
GET | /matters/{id}/evidence | Evidence list
POST | /matters/{id}/documents/upload | Upload to matter
GET | /matters/{id}/contradictions | Contradiction report

4.4 Documents & KB API

Method | Path | Purpose

POST | /documents/upload | Upload + index PDF
GET | /documents/kb/health | Index health
POST | /documents/kb/reindex-auto | Trigger reindex
GET | /documents/jobs/{job_id} | Index job status

4.5 Evidence Intelligence API

Method | Path | Purpose

POST | /ediscovery/evidence/upload | Multipart file upload
GET | /ediscovery/evidence/repository | Matter evidence list
GET | /ediscovery/evidence/timeline | Merged timeline
POST | /ediscovery/evidence/contradictions | Compare two texts
POST | /ediscovery/evidence/statutes | BNS/BNSS suggestions
POST | /ediscovery/evidence/court-orders | Similar orders search

4.6 Drafting API (Selected)

Method | Path | Purpose

GET | /drafting/workspace/documents | List drafts
POST | /drafting/workspace/documents | Create draft
PATCH | /drafting/workspace/documents/{id}/content | Save content
POST | /drafting/workspace/documents/{id}/export |

4.7 Learning / Feedback API

POST /api/v1/learning/feedback

Authorization: Bearer <token>

```
{
  "signal": "thumbs_up",
  "interaction_id": "uuid-from-chat-response",
  "chat_id": "optional"
}
```

4.8 IPC-BNS Legal Tools API

Method | Path | Purpose

GET | /ipc-bns/v3/search?q=cheating | Search mappings

GET | /ipc-bns/v3/compare/{ipc_section} | IPC vs BNS compare

POST | /ipc-bns/v3/bulk | Bulk conversion

POST | /ipc-bns/v3/document/upload | Upload + convert

4.9 Enterprise Workspace API

Method | Path | Purpose

GET | /enterprise/workspace/dashboard | Firm dashboard

POST | /enterprise/workspace/court-orders | Upload order

GET | /enterprise/workspace/court-orders/search | Search orders

POST | /enterprise/workspace/knowledge | Create KB entry

4.10 Complete Route Count

The API exposes **503 HTTP routes + 1 WebSocket** across 38 endpoint modules. Full catalog maintained in backend/app/api/v1/router.py and generated from endpoint files at build time.

Document 5 - AI Architecture

5.1 Design Principles

1. ****Grounded over generative**** - KB answers must cite indexed documents
2. ****Mode isolation**** - KB uses Ollama; web uses Gemini; never cross-contaminate
3. ****Feedback-driven improvement**** - Thumbs adjust retrieval weights, not model weights live
4. ****Graceful degradation**** - web_provider_chain falls through 6 providers to local Ollama
5. ****Matter isolation**** - Separate FAISS indexes prevent cross-case leakage

5.2 Knowledge Base RAG Pipeline

```
Upload PDF
  |
  v
OCR Gate (150 chars/page threshold)
  |
  v
Text Extraction (PyMuPDF / pdfplumber / EasyOCR)
  |
  v
Chunking (RAG_CHUNK_SIZE, overlap)
  |
  v
Embeddings (HF sentence-transformers)
  |
  v
FAISS Index (per-user global or matter-scoped)
  |
  v
Query --> Hybrid Retrieve (dense + sparse + MMR)
  |
  v
Cross-encoder Rerank (optional)
  |
  v
kb_pipeline: Intent --> Filter --> Aggregate
  |
  v
Ollama LLM (legalease-tuned) --> Validated Answer
```

****Key files:**** rag.py, kb_pipeline.py,
kb_retrieval_coordinator.py, llms.py

5.3 Open Law Pipeline

```
Question
|
v
Research Dimension Detection (web_intelligence.py)
|
v
Gemini Grounded Search (Google Search tool)
|
v (fallback)
Tavily / SerpAPI / DuckDuckGo
|
v (fallback)
OpenRouter / DeepSeek / Qwen
|
v (fallback)
Local Ollama legal_reason()
|
v
Formatted Answer + web_sources[]
```

5.4 Hybrid Mode

```
Query
|
+-- KB Retrieve (top-k chunks from matter or global)
|
+-- Web Research (Gemini)
|
v
Fusion (hybrid_orchestrator.py)
|
v
Unified answer with KB citations + web sources
```

5.5 Adaptive Learning Loop

```
Chat Turn --> record_interaction (adaptive_interactions)
|
v
User Thumbs Up/Down --> record_feedback
|
v
Update chunk boosts + query patterns (adaptive_chunk_boosts, adaptive_query_patterns)
|
v
Optional: Gemini-Ollama Coach (offline settings tuning)
|
v
Optional: Export JSONL for external LoRA fine-tuning
```

5.6 Evidence Intelligence AI

Rule-based + pattern classification
(production-safe without GPU):

- Document category classification (regex + keyword rules)

- Entity extraction (emails, phones, IFSC, case numbers, dates, org names)

- Risk indicators (fraud, bribery, conspiracy patterns)

- Privilege markers (attorney-client, work-product)

- Statute mapping (BNS 316, 318, etc.)

- Contradiction detection (negation conflicts, date/amount mismatches)

Future: LLM enrichment layer for complex privilege review (roadmap).

5.7 LLM Backend Selection

Environment | LLM_BACKEND | KB LLM | Web LLM

Laptop | ollama | Ollama GPU | Ollama or Gemini

EC2 Production | gemini | Ollama CPU fallback |

Gemini primary

Document 6 - Security Architecture

6.1 Authentication

```
Login --> bcrypt password verify (legalease_auth.py)
      |
      v
JWT signed with HMAC-SHA256 (JWT_SECRET / LEGALEASE_API_SECRET)
      |
      v
Token TTL: LEGALEASE_TOKEN_TTL (default 24h)
      |
      v
Each request: decode_access_token --> get_current_user
      |
      v
Live membership refresh from DB (plan changes immediate)
```

Optional: OIDC SSO via /api/v1/sso
(sso_service.py)

6.2 Authorization Roles

Role | Scope

user | Standard attorney access

admin / superadmin | Admin panel, user suspension,
metrics

org owner | Team invites, branding, billing

org lawyer | Matter write access

org member/viewer | Read-only matter access

client (portal) | Token-scoped read/upload only

6.3 Data Protection

Passwords: bcrypt (BYTEA on Postgres -
fixed for EC2)

Optional field encryption:

DATA_ENCRYPTION_KEY (Fernet)

HTTPS: nginx TLS termination, HSTS
headers

CORS: CORS_ORIGINS whitelist in
production

Rate limiting: RATE_LIMIT_PER_MINUTE,
chat-specific limits

IP firewall: FIREWALL_ENABLED optional

allowlist

PII redaction:

document_services/pii_redactor.py

Audit logs: audit_events,

matter_audit_log, ent_audit

6.4 Multi-Tenant Isolation

All queries scoped by user_id from JWT

Matter access enforced by

matter_policy.py

FAISS indexes physically separated per

user/matter

Org seat limits enforced by

plan_enforcement.py

6.5 Security Headers (Middleware Stack)

1. Memory guard

2. Rate limit

3. Request guard

4. IP firewall (optional)

5. Security headers (X-Frame-Options, CSP
baseline)

6. CORS

Reference: SECURITY.md,
</api/v1/health/security>

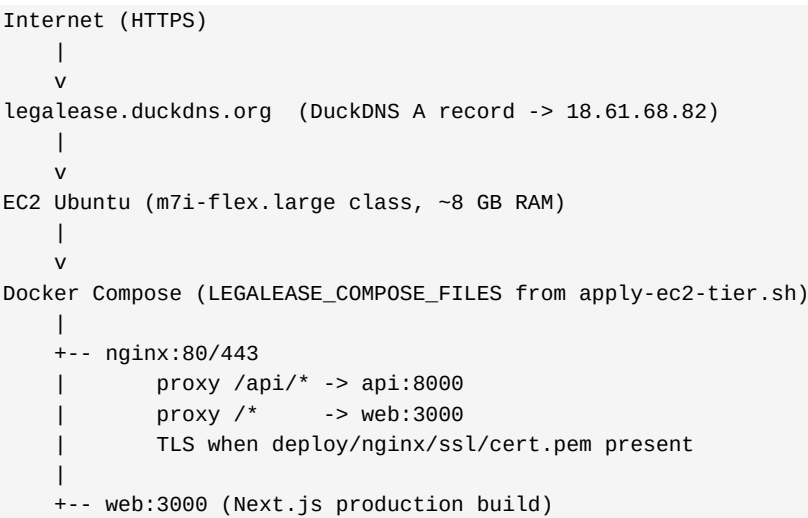
Document 7 - Deployment Architecture (Complete Guide)

This section is the authoritative deployment reference for LegalEase.AI. Every command, URL, script, and environment profile below uses the actual project configuration. Follow it end-to-end without adding external notes.

7.1 Production Summary

Item	Value
Public URL	https://legalease.duckdns.org
API base	https://legalease.duckdns.org/api
Health check	https://legalease.duckdns.org/api/v1/health/live
EC2 public IP	18.61.68.82
EC2 SSH user	ubuntu
Server install path	/opt/legalease
SSH key (Windows)	%USERPROFILE%\ssh\legalease-aws.pem
Stack	Docker Compose: nginx + web + api + postgres + redis
LLM (production)	Gemini (CLOUD_GEMINI_KB=1, LLM_BACKEND=gemini)
Database (production)	PostgreSQL 16 in Docker
Sessions	Redis 7
TLS	DuckDNS + optional Let's Encrypt certs in deploy/nginx/ssl/

7.2 Architecture Diagram (Production)



```
|      NEXT_PUBLIC_API_URL=https://legalease.duckdns.org/api (baked at build time)
|
+-- api:8000 (FastAPI, UVICORN_WORKERS=1 on low RAM)
|      Gemini for Open Law / Hybrid web leg
|      FAISS indexes on /data/faiss_indexes volume
|
+-- postgres:5432 (volume postgres_data)
+-- redis:6379 (volume redis_data, appendonly)
|
+-- worker / ml-worker (profile: workers - OFF by default on 8GB)
```

7.3 Laptop Development - Step-by-Step Setup

Prerequisites (Windows laptop)

1. Install Python 3.10+ and Node.js 18+
2. Install Ollama from <https://ollama.com> and run `ollama pull legalease-tuned` (or your tuned model)
3. Install ffmpeg: `winget install Gyan.FFMpeg`
4. Optional GPU: NVIDIA drivers + CUDA for Ollama GPU and faster embeddings
5. Clone or open project folder:
`Legal\_AI\_Final 3`

One-time laptop configuration

```
cd "C:\Users\ASUS\Desktop\Legal_ai (1)\Legal_ai\Legal_AI_Final 3"

# API keys (Gemini, Tavily) go in .env - copy from .env.example if missing
# Laptop-only overrides:
.\scripts\setup_local_env.ps1
# Creates .env.local from .env.local.example with:
#   LEGALEEASE_LOCAL_DEV=1
#   SAAS_PRODUCTION=0
#   SAAS_USE_POSTGRES_LEGACY=0   (SQLite)
#   LLM_BACKEND=ollama
#   NEXT_PUBLIC_API_URL=http://127.0.0.1:8000
#   CORS_ORIGINS=http://localhost:3000,http://127.0.0.1:3000
```

Edit `.env` for Ollama settings (already in `.env.example`):

```
LLM_BACKEND=ollama
OLLAMA_BASE_URL=http://127.0.0.1:11434
OLLAMA_MODEL=legalease-tuned
CLOUD_GEMINI_KB=0
GEMINI_API_KEY=your-key-for-open-law-only
```

Create Python venv (first time only):

```
py -m venv .venv_win
.\venv_win\Scripts\pip install -r backend\requirements.txt
```

```
cd web; npm install; cd ..
```

Start laptop stack (every session)

Terminal 1 - Backend:

```
.\run_backend.ps1
```

Terminal 2 - Frontend:

```
.\run_web.ps1
```

Open browser: <http://localhost:3000>

Health:

<http://127.0.0.1:8000/api/v1/health/live>

API docs: <http://127.0.0.1:8000/docs>

What run_backend.ps1 does automatically

Calls `\scripts\apply_local_env.ps1`` to
merge `.env.local`` overrides

Sets `LEGALEASE_DB_PATH`` to project
`legalease.db`` (SQLite)

Sets `SAAS_PRODUCTION=0``, skips blocking
RAG warmup for fast boot

Auto-starts Ollama on GPU when
`OLLAMA_AUTO_START=1``

Runs uvicorn on 127.0.0.1:8000 with 300s
keep-alive

Laptop vs EC2 separation (critical)

File | Used on | Must NOT contain

`.env` | Both (API keys) | `SAAS_PRODUCTION=1` on laptop

`.env.local` | Laptop only (gitignored) |
postgres/redis Docker hostnames

`/opt/legalease/.env` | EC2 only | localhost database
URLs

`aws_update.ps1`` explicitly excludes

`.env`` and `.env.local`` from upload.

Server `.env`` is never overwritten by
deploy.

7.4 EC2 - Initial Server Setup (First Time)

7.4.1 AWS resources

1. Launch Ubuntu 22.04+ EC2 instance
(recommended: m7i-flex.large, 8 GB RAM)
2. Security group inbound rules:
SSH 22 from your IP
HTTP 80 from 0.0.0.0/0 (required for DuckDNS)
HTTPS 443 from 0.0.0.0/0 (when TLS certs mounted)
3. Elastic IP optional; current
production IP: 18.61.68.82
4. DuckDNS: point `legalease.duckdns.org`
A record to 18.61.68.82

See `deploy/aws/OPEN\_PORT\_80.md` if
DuckDNS times out (port 80 blocked).

7.4.2 SSH access from Windows

```
ssh -i $env:USERPROFILE\.ssh\legalease-aws.pem ubuntu@18.61.68.82
```

7.4.3 Install Docker on EC2 (if fresh server)

```
sudo apt-get update
sudo apt-get install -y docker.io docker-compose-v2 git
sudo usermod -aG docker ubuntu
# Log out and back in
```

7.4.4 First deploy from Windows

From project root on your laptop:

```
.\scripts\aws_update.ps1 -VmIp 18.61.68.82 -PublicUrl "https://legalease.duckdns.org"
```

This script:

1. Creates `legalease-deploy.tgz`
(excludes Data, node_modules, .next, venv, .env)
2. SCP upload to
`/opt/legalease/legalease-deploy.tgz`
3. SSH runs remote script that:
Extracts tarball to `/opt/legalease`
Runs `deploy/aws/fix-ec2-env.sh`

```
'https://legalease.duckdns.org'
```

```
Runs `deploy/aws/fix-postgres-  
password.sh`
```

```
Runs `deploy/aws/ec2-go-live.sh`  
'https://legalease.duckdns.org'
```

4. Rebuilds api + web Docker images with
correct `NEXT\_PUBLIC\_API\_URL`

5. Starts nginx, api, web, postgres,
redis

Expected duration: 5-15 minutes.

7.4.5 Create production .env on server (first time only)

SSH to EC2 and create

`/opt/legalease/.env` from template:

```
cd /opt/legalease  
cp deploy/aws/.env.production.example .env  
nano .env
```

Required values to set (generate secrets
with `pwsh scripts/rotate\_secrets.ps1` on
laptop):

```
POSTGRES_PASSWORD=<strong-random-password>  
JWT_SECRET=<32+-char-random>  
LEGALEASE_API_SECRET=<32+-char-random>  
DATA_ENCRYPTION_KEY=<32+-char-random>  
GEMINI_API_KEY=<your-google-ai-key>  
PUBLIC_APP_URL=https://legalease.duckdns.org  
CORS_ORIGINS=https://legalease.duckdns.org  
NEXT_PUBLIC_API_URL=https://legalease.duckdns.org/api  
DATABASE_URL=postgresql://legalease:<POSTGRES_PASSWORD>@postgres:5432/legalease  
REDIS_URL=redis://redis:6379/0  
SAAS_USE_POSTGRES_LEGACY=1  
SAAS_PRODUCTION=1  
LLM_BACKEND=gemini  
CLOUD_GEMINI_KB=1
```

Stripe (before SAAS_PRODUCTION_STRICT=1):

```
STRIPE_SECRET_KEY=sk_live_...  
STRIPE_WEBHOOK_SECRET=whsec_...  
STRIPE_PRICE_PRO=price_...  
STRIPE_PRICE_LEGAL_PRO=price_...
```

Then re-run go-live:

```
bash deploy/aws/ec2-go-live.sh https://legalease.duckdns.org
```

7.5 EC2 - Re-Deploy After Code Changes

Full update (recommended)

```
cd "C:\Users\ASUS\Desktop\Legal_ai (1)\Legal_ai\Legal_AI_Final 3"  
.\scripts\aws_update.ps1 -VmIp 18.61.68.82 -PublicUrl "https://legalease.duckdns.org"
```

Hotfix only (5 deploy files, no full tarball)

```
.\scripts\aws_go_live.ps1 -VmIp 18.61.68.82 -PublicUrl "https://legalease.duckdns.org"
```

Post-deploy verification checklist

1. ``curl https://legalease.duckdns.org/api/v1/health/live`` returns 200
2. ``curl https://legalease.duckdns.org/api/v1/health/public`` shows
``core_db.backend: postgresql``
3. Open <https://legalease.duckdns.org> in browser - hard refresh (Ctrl+Shift+R)
4. Login as admin user
5. Test chat (KB mode) and thumbs feedback
6. Test Evidence Intelligence upload on /discovery

7.6 Docker Compose Services Reference

Service | Image/Build | Ports | Memory limit (EC2 low) | Role

postgres | postgres:16-alpine | 5432 internal | default | All SaaS tables when SAAS_USE_POSTGRES_LEGACY=1

redis | redis:7-alpine | 6379 internal | default | Chat sessions, ML job queues

api | deploy/Dockerfile.api.aws | 8000 internal | 4G | FastAPI REST + SSE

web | deploy/Dockerfile.web | 3000 internal | 1536M | Next.js UI

nginx | nginx:alpine | 80, 443 | default | Reverse proxy

worker | same as api | - | profile workers | E-discovery job processor

ml-worker | same as api | - | profile workers | ML tuning / reindex jobs

Compose file stack (from

`deploy/aws/apply-ec2-tier.sh`):

```
-f docker-compose.yml
-f deploy/aws/docker-compose.override.yml
# + deploy/aws/docker-compose.highmem.yml when RAM tier != low
# + deploy/aws/docker-compose.https.yml when SSL certs exist
```

Environment exported to `/tmp/legalease-compose.env`:

```
LEGALEASE_COMPOSE_FILES="-f docker-compose.yml -f deploy/aws/docker-compose.override.yml ..."
eval "$(cat /tmp/legalease-compose.env)"
docker compose ${LEGALEASE_COMPOSE_FILES} ps
```

7.7 EC2 Memory Tiers (apply-ec2-tier.sh)

Tier | Detection | ML_USE_QUEUE | LOW_RESOURCE_MODE | STT model | Workers

low | <=8 GB | 0 | 1 | tiny | disabled

medium | 8-16 GB | 1 | 0 | small | optional

high | >16 GB | 1 | 0 | small | can enable

On low tier (current 8GB production):

```
`UVICORN_WORKERS=1`
`RAG_ENABLE_CROSS_ENCODER=0`
`HF_EMBEDDING_MODEL=BAAI/bge-small-en-v1.5`

Ollama disabled; Gemini required for web
modes
```

Enable workers on high-memory instance:

```
docker compose $LEGALEASE_COMPOSE_FILES --profile workers up -d worker ml-worker
```

7.8 nginx Configuration

Production nginx config:

`deploy/nginx/nginx.conf` (HTTP) and
`deploy/nginx/nginx-ssl.conf` (HTTPS).

Key settings:

```
`/api/` proxied to `http://api:8000/api`
`/` proxied to `http://web:3000/`
`proxy_read_timeout 300s` for long chat
streams

Rate limiting zones for auth and chat
endpoints
```

TLS certificates path:


```
`deploy/nginx/ssl/cert.pem` (full chain)
`deploy/nginx/ssl/key.pem` (private key)
```

When both exist, `apply-ec2-tier.sh` adds
`-f deploy/aws/docker-compose.https.yml`.

7.9 DuckDNS and TLS Setup

DuckDNS

1. Register subdomain at duckdns.org
2. Set A record to 18.61.68.82
3. Ensure AWS security group allows inbound TCP 80
4. Set in server `.env`:
`PUBLIC\_APP\_URL=https://legalease.duckdns.org`
`CORS\_ORIGINS=https://legalease.duckdns.org`
`NEXT\_PUBLIC\_API\_URL=https://legalease.duckdns.org/api`

Let's Encrypt (optional HTTPS)

On EC2 with port 443 open:

```
sudo apt install certbot
sudo certbot certonly --standalone -d legalease.duckdns.org
sudo cp /etc/letsencrypt/live/legalease.duckdns.org/fullchain.pem deploy/nginx/ssl/cert.pem
sudo cp /etc/letsencrypt/live/legalease.duckdns.org/privkey.pem deploy/nginx/ssl/key.pem
bash deploy/aws/apply-ec2-tier.sh
bash deploy/aws/ec2-go-live.sh https://legalease.duckdns.org
```

Cloudflare quick tunnel (fallback when port 80 blocked)

`ec2-go-live.sh` auto-starts cloudflared
if direct HTTP fails:

```
cloudflared tunnel --url http://127.0.0.1:80
# URL logged to /tmp/cloudflared.log
# systemd unit: cloudflared-legalease.service
```

Use tunnel URL as `PublicUrl` until port 80 is opened.

7.10 Environment Variable Profiles (Laptop vs EC2)

Variable | Laptop (.env.local) | EC2

(/opt/legalease/.env)

LEGALEEASE_LOCAL_DEV | 1 | unset
SAAS_PRODUCTION | 0 | 1
SAAS_PRODUCTION_STRICT | 0 | 1 (after Stripe keys set)
SAAS_USE_POSTGRES_LEGACY | 0 | 1
DATABASE_URL | empty (SQLite) | postgresql://legalease:PASS@postgres:5432/legalease
LEGALEASE_DB_PATH | legalease.db | /data/legalease.db
REDIS_URL | optional | redis://redis:6379/0
LLM_BACKEND | ollama | gemini
CLOUD_GEMINI_KB | 0 | 1
OLLAMA_AUTO_START | 1 | 0
NEXT_PUBLIC_API_URL | http://127.0.0.1:8000 | https://legalease.duckdns.org/api
CORS_ORIGINS | http://localhost:3000 | https://legalease.duckdns.org
PUBLIC_APP_URL | http://localhost:3000 | https://legalease.duckdns.org
ALLOW MOCK BILLING | 1 | 0
SAAS_ALL_FEATURES_FREE | 1 | 1 (demo)
RATE_LIMIT_ENABLED | 0 or 1 | 1
STT_ENABLED | 1 | 1
STT_DEVICE | cuda if GPU | cpu
STT_MODEL | base/small | tiny (low RAM)
IMPROVEMENT_AUTO | 1 | 0
COACH_AUTO_SCHEDULE | 1 | 0

7.11 Stripe Billing Setup (Production)

1. Create Stripe account and products (Pro, Legal Pro)
2. Copy price IDs to `.env``:
``STRIPE_PRICE_PRO``,
``STRIPE_PRICE_LEGAL_PRO``
3. Set ``STRIPE_SECRET_KEY`` and ``STRIPE_WEBHOOK_SECRET``
4. Configure Stripe webhook endpoint: ``https://legalease.duckdns.org/api/v1/subscriptions/webhook``
5. Set ``SAAS_PRODUCTION_STRICT=1`` only after keys are live

6. If API unhealthy due to placeholder
Stripe key, temporarily set
`SAAS\_PRODUCTION\_STRICT=0`

7.12 Database Backup and Restore

Backup (production)

```
ssh -i ~/.ssh/legalease-aws.pem ubuntu@18.61.68.82
cd /opt/legalease
export DATABASE_URL=postgresql://legalease:YOUR_PASS@postgres:5432/legalease
docker compose $LEGALEASE_COMPOSE_FILES exec -T postgres pg_dump -U legalease legalease >
backups/postgres_$(date +%Y%m%)
```

Or from laptop:

```
py scripts\backup_legalease.py --out backups/manual
```

Backup includes: postgres dump, SQLite (if present), faiss_indexes/, Data/

Restore procedure

1. Stop API: `docker compose \$LEGALEASE\_COMPOSE\_FILES stop api web`
2. Restore Postgres: `pg\_restore -d legalease backups/postgres\_YYYYMMDD.dump`
3. Restore files: copy faiss_indexes/ and Data/ from backup
4. Start stack: `bash deploy/aws/ec2-go-live.sh https://legalease.duckdns.org`
5. Verify: `curl https://legalease.duckdns.org/api/v1/health/public`

SQLite to Postgres migration (one-time)

```
py scripts\migrate_sqlite_to_pg.py
```

Run when moving from laptop SQLite to Docker Postgres.

7.13 Rollback Procedure

If deploy breaks production:

```
ssh ubuntu@18.61.68.82
cd /opt/legalease
# View previous images
docker images | grep legalease
```

```
# Roll back to previous git state if tagged
git log --oneline -5
# Or restore from backup (section 7.12)
docker compose $LEGALEASE_COMPOSE_FILES logs api --tail 100
```

From Windows, redeploy last known good commit:

```
git checkout <last-good-commit>
.\scripts\aws_update.ps1 -PublicUrl "https://legalease.duckdns.org"
```

7.14 Monitoring and Logs

```
# All services
docker compose $LEGALEASE_COMPOSE_FILES ps
docker compose $LEGALEASE_COMPOSE_FILES logs -f api
docker compose $LEGALEASE_COMPOSE_FILES logs -f web
docker compose $LEGALEASE_COMPOSE_FILES logs -f nginx

# Health endpoints
curl https://legalease.duckdns.org/api/v1/health/live
curl https://legalease.duckdns.org/api/v1/health/ready
curl https://legalease.duckdns.org/api/v1/health/public
curl https://legalease.duckdns.org/api/v1/health/llm
```

Structured logs in API include request ID, user ID, and pipeline stage when `KB\_PIPELINE\_DEBUG=1`.

7.15 CI/CD (GitHub Actions)

File: `.github/workflows/ci.yml`

On every push/PR:

- Python pytest (~126 tests)
- Next.js production build verification
- Lint checks

CI does not auto-deploy to EC2. Production deploy is manual via `aws\_update.ps1`.

7.16 Troubleshooting Matrix (Complete)

Symptom | Likely cause | Fix

502 Bad Gateway on /api | API container down or starting | `docker compose logs api`; wait 90s start_period

CORS error in browser | CORS_ORIGINS mismatch | Set CORS_ORIGINS exactly to https://legalease.duckdns.org (no trailing slash)

Web shows wrong API | Stale NEXT_PUBLIC_API_URL |

Rebuild web: `ec2-go-live.sh`
<https://legalease.duckdns.org>
 password authentication failed for postgres |
 POSTGRES_PASSWORD drift | `bash deploy/aws/fix-postgres-password.sh`
 STRIPE_SECRET_KEY placeholder error |
 SAAS_PRODUCTION_STRICT=1 without keys | Add Stripe keys or set STRICT=0 temporarily
 Thumbs feedback not saving | adaptive_mode_stats schema | Deploy latest code (pg_core_schema migration)
 KB returns NOT_FOUND | Documents not indexed | Upload PDFs, click Index All, check faiss_indexes volume
 Open Law quota exceeded | Gemini daily limit | Wait for reset or upgrade plan; Tavily fallback
 Ollama connection refused on EC2 | Expected - EC2 uses Gemini | Set LLM_BACKEND=gemini, CLOUD_GEMINI_KB=1
 Port 8000 in use on laptop | Stale Python process |
 `.\stop\_backend.ps1` then `run\_backend.ps1`
 DuckDNS timeout | Port 80 blocked | Open AWS SG port 80 or use Cloudflare tunnel
 Web build fails premium/page | Stale route |
 aws_update.ps1 removes web/app/(app)/premium
 Sessions lost between requests | Redis not configured | Set REDIS_URL=redis://redis:6379/0
 OCR fails on scans | EasyOCR not in container |
 Rebuild api image; check OCR_ENABLED=1
 STT fails in browser | ffmpeg missing on laptop |
 winget install Gyan.FFmpeg
 Feedback 500 on Postgres | Transaction rollback on stats | Latest adaptive_learning.py commits interaction first
 Index job stuck | ML_USE_QUEUE=0 on low RAM | Run index from UI; or enable worker profile
 High API memory | FAISS + embeddings |
 LOW_RESOURCE_MODE=1, RAG_ENABLE_CROSS_ENCODER=0
 TLS handshake error | Cert path wrong | Verify
 deploy/nginx/ssl/cert.pem and key.pem
 Login works locally not EC2 | Wrong DATABASE_URL |
 Must use @postgres:5432 not @localhost
 Deploy tarball huge | Data included | aws_update excludes Data, faiss_indexes, node_modules

7.17 fix-ec2-env.sh - Line-by-Line Behavior

Script: `deploy/aws/fix-ec2-env.sh`

1. Sets working directory to
`/opt/legalease`
2. Accepts public URL argument (defaults to <http://18.61.68.82>)

3. Forces SAAS_USE_POSTGRES_LEGACY=1
4. Runs apply-ec2-tier.sh for memory tier compose files
5. Rewrites DATABASE_URL localhost to postgres hostname
6. Rewrites REDIS_URL 127.0.0.1 to redis hostname
7. Sets CLOUD_GEMINI_KB=1 if missing
8. Sets CORS_ORIGINS, PUBLIC_APP_URL, NEXT_PUBLIC_API_URL from public URL
9. Configures CPU speech-to-text (STT_MODEL=tiny, STT_DEVICE=cpu)
10. Rebuilds api + web containers
11. Restarts api, web, nginx

7.18 ec2-go-live.sh - Line-by-Line Behavior

Script: `deploy/aws/ec2-go-live.sh`

1. Detects EC2 public IP via checkip.amazonaws.com
2. Sets SAAS_USE_POSTGRES_LEGACY=1, SAAS_PRODUCTION=1
3. Configures STT for CPU
4. Runs apply-ec2-tier.sh
5. `docker compose up -d --build` full stack
6. If no PUBLIC_BASE arg and port 80 unreachable: starts Cloudflare tunnel
7. Writes systemd unit for cloudflared persistence
8. Updates CORS, PUBLIC_APP_URL, NEXT_PUBLIC_API_URL
9. Rebuilds api + web with exported NEXT_PUBLIC_API_URL
10. Health check loop (60 attempts, 2s interval)
11. Prints live URL and health response

7.19 Docker Local Production (Non-EC2)

For self-hosted Docker on any server:

```
copy .env.docker.example .env
# Edit POSTGRES_PASSWORD, JWT_SECRET, NEXT_PUBLIC_API_URL, CORS_ORIGINS
# Place TLS certs in deploy/nginx/ssl/
docker compose up -d --build
```

Health:

<http://localhost/api/v1/health/live>

See DEPLOY.md for LM Studio on Windows

host (`LM\_STUDIO\_URL=http://host.docker.internal:1234`).

7.20 Post-Deploy Smoke Tests

```
# From laptop
curl https://legalease.duckdns.org/api/v1/health/live
py scripts\e2e_saas_smoke.py --url https://legalease.duckdns.org/api
py scripts\e2e_kb_smoke.py --url https://legalease.duckdns.org/api
```

Manual UI tests:

1. Register / login
2. Upload PDF to Documents, Index All
3. KB chat with citation
4. Create matter, link document
5. Evidence Intelligence upload on /discovery
6. Thumbs up on chat response - verify POST /api/v1/learning/feedback returns ok:true
7. Billing invoice PDF export (if Stripe configured)

7.21 Security Hardening Checklist

- [] JWT_SECRET and LEGALEASE_API_SECRET are unique 32+ char random strings
- [] POSTGRES_PASSWORD is strong and not committed to git
- [] GEMINI_API_KEY only on server .env
- [] SAAS_PRODUCTION_STRICT=1 after Stripe configured

- [] FORCE_HTTPS=1 and
SECURITY_HEADERS_ENABLED=1 on production
- [] FIREWALL_ENABLED=0 or restrict
FIREWALL_ALLOWED_IPS for admin
- [] SUPERADMIN_USERNAMES lists only
trusted admins
- [] Regular Postgres backups scheduled
- [] SSH key-only access (disable password
auth)
- [] AWS security group limits SSH to your
IP

7.22 Volume and Data Persistence

Volume | Mount | Contents

postgres_data | postgres | All PostgreSQL data
redis_data | redis | Session/cache AOF
app_data | api | /data/legalease.db fallback, uploads
./faiss_indexes | api | Per-user FAISS vector indexes
./Data | api | Uploaded PDFs and HF cache

Never delete postgres_data without backup.
faiss_indexes can be rebuilt via reindex
but takes time.

Document 8 - Product Workflow Guide

8.1 Matter Lifecycle

```
Create Matter
  |
  v
Assign Lawyer + Set Practice Area
  |
  v
Upload Documents (global KB or matter-scoped)
  |
  v
Index (FAISS) -- automatic or job queue
  |
  v
AI Research (KB / Hybrid / Open Law)
  |
  v
Draft Pleadings (Drafting Studio)
  |
  v
Track Hearings (Litigation Desk / cause list import)
  |
  v
Evidence Review (Evidence Intelligence Center)
  |
  v
Billing + Invoice
  |
  v
Close / Archive Matter
```

8.2 Client Portal Workflow

```
Lawyer generates portal link (POST /portal/access)
  |
  v
Client opens magic URL (/portal/{token})
  |
  v
View matter status + timeline (read-only)
  |
  v
Upload requested documents (POST /portal/upload/{token})
  |
  v
Review / approve drafts (enterprise client portal)
  |
  v
Download court orders (when shared)
```

8.3 Evidence Investigation Workflow

```
Select Matter
|
v
Upload evidence files (drag-and-drop)
|
v
Automatic: OCR + Metadata + Classification
|
v
Review Evidence Strength + Privilege flags
|
v
Inspect Entity cards + Timeline
|
v
Run Contradiction Check (2 documents)
|
v
Statute Finder (BNS sections)
|
v
Court Order Matcher (firm KB)
|
v
Evidence stored in Repository (linked to matter)
```

8.4 Frontend Routes Reference

Route | Module

```
/ | AI Assistant
/dashboard | Firm dashboard
/matters | Matter list
/matters/{id}/timeline | Case timeline
/matters/{id}/evidence | Matter evidence
/discovery | Evidence Intelligence Center
/drafting | Drafting Studio
/tools/ipc-bns | IPC-BNS converter
/intake | CRM Intake Desk
/billing | Billing
/litigation | Litigation Desk
/enterprise | Enterprise workspace
/collaboration | Firm Chat
/analytics | Analytics
/settings | Settings + Memory
```

Document 9 - Competitive Analysis

9.1 Market Landscape

Competitor | Category | Strength | Gap vs LegalEase

Harvey AI | AI legal assistant | Brand, LLM quality |
No Indian IPC-BNS, no integrated DMS

Relativity | E-discovery | Enterprise scale |
Expensive, not AI-native for Indian law

Everlaw | E-discovery | Review workflows | US-
focused, no statute tools

Clio | Practice management | Market leader PM | Weak
AI, no RAG

MyCase | PM + client portal | SMB friendly | No
evidence intelligence

LexisNexis | Research | Authority content |
Expensive, not matter-integrated AI

Westlaw | Research | Case law depth | Same as Lexis

DISCO | E-discovery | Cloud review | No Indian legal
tools

Logikcull | E-discovery | Simple upload | No
drafting/KB

9.2 Feature Comparison Matrix

Feature | LegalEase | Harvey | Relativity | Clio | LexisNexis

Document RAG / KB | Yes | Partial | No | No | No

Open Law web research | Yes | Yes | No | No | Yes

IPC to BNS converter | Yes | No | No | No | Partial

Evidence Intelligence | Yes | No | Yes | No | No

Drafting Studio | Yes | Yes | No | Basic | No

Matter workspace | Yes | Partial | Yes | Yes | No

CRM Intake | Yes | No | No | Yes | No

Billing + Trust | Yes | No | No | Yes | No

Indian cause lists | Yes | No | No | No | No

Adaptive learning | Yes | Unknown | No | No | No

Client portal | Yes | No | No | Yes | No

Price (SMB) | Low SaaS | Enterprise | Enterprise |
Mid | High

9.3 LegalEase Unique Value Proposition

1. ****India-first**** - BNS/BNSS, IPC migration, cause lists, eCourts integration path
 2. ****Unified stack**** - One login for AI + matters + evidence + drafting + billing
 3. ****Evidence Intelligence Center**** - Relativity-style investigation without enterprise pricing
 4. ****Self-improving KB**** - Feedback loop without per-query GPU fine-tuning
 5. ****Deploy flexibility**** - Laptop GPU for dev, cloud Gemini for production
-

Document 10 - Technical Deep Dive

10.1 Why FastAPI?

Native async for SSE chat streaming and concurrent uploads

Pydantic validation on 503 endpoints

Automatic OpenAPI docs for developer onboarding

Python ecosystem access to PyMuPDF, FAISS, sentence-transformers, bcrypt

Middleware stack for rate limiting and security headers

10.2 Why PostgreSQL (Production)?

ACID transactions for billing, trust ledger, evidence chain-of-custody

Concurrent writes from multiple API workers

JSON columns for metadata_json, entities_json on discovery_items

Mature backup/restore on EC2 Docker volumes

SQLite retained for zero-config laptop development

10.3 Why RAG (Retrieval-Augmented Generation)?

Legal answers require ****citations**** - pure LLM hallucinates sections

Attorneys trust answers tied to ****their**** uploaded documents

RAG allows NOT_FOUND when confidence below threshold (honest system)

Chunk-level feedback enables targeted retrieval improvement

10.4 Why Vector Search (FAISS)?

Semantic search finds relevant passages even when wording differs

Hybrid dense+sparse+MMR reduces missed sections in long statutes

Per-matter indexes enforce confidentiality boundaries

CPU-FAISS sufficient for SMB firm document volumes

10.5 Why OCR?

Indian courts and clients deliver scanned PDFs without text layer

Evidence Intelligence requires text from images, WhatsApp screenshots

150 chars/page gate avoids unnecessary OCR on digital PDFs

EasyOCR + Tesseract fallback chain in `ocr_router.py`

10.6 Why Role-Based Access?

Law firms require partner/associate/paralegal separation

Client portal must be read-only with token expiry

Matter-level write access prevents junior staff editing senior cases

Admin role for billing and user suspension

10.7 Why Audit Logs?

Legal ethics require traceability on document access

Enterprise sales require audit trail demonstrations

`matter_audit_log`, `ent_audit`, `collab_audit` support compliance story

Feedback and learning signals auditable
for model governance

10.8 Why Separate Laptop vs EC2 Config?

Developers need GPU Ollama locally for
fast KB iteration

EC2 8GB RAM cannot run GPU Ollama at
scale - Gemini for web

.env.local gitignored prevents accidental
production DB connection

apply_local_env.ps1 enforces
SAAS_PRODUCTION=0 on laptop

10.9 Testing & Quality

~126 pytest tests: KB, RAG, memory, auth,
phase4 SaaS

KB smoke tests via /documents/kb/smoke-
test

CI on GitHub Actions for every PR

Health endpoints for production
monitoring

10.10 Key File Reference

Concern | Primary Files

Chat | backend/app/services/chat_service.py

RAG | rag.py, kb_pipeline.py

Auth | legacy_saas/legalease_auth.py,
backend/app/core/auth.py

Matters | backend/app/core/matter_repo.py

Evidence | backend/app/core/evidence_intelligence.py

Deploy | docker-compose.yml, scripts/aws_update.ps1

Frontend | web/app/(app)/, web/lib/api.ts

Product Roadmap & Appendix

Phase 1 (Complete) - Core Platform

KB chat, matters, documents, auth, billing
foundation

Phase 2 (Complete) - Practice Ops

CRM intake, billing invoices, trust
ledger, collaboration

Phase 3 (Complete) - Drafting & Tools

Drafting Studio v4, IPC-BNS engine,
litigation desk

Phase 4 (Complete) - Evidence & Research

Evidence Intelligence Center, adaptive
learning, feedback loops

Phase 5 (Complete) - Enterprise

DMS, court orders, knowledge base, client
portal, analytics

Phase 6 (Roadmap)

Live eCourts API deep integration
LLM-based privilege review in Evidence
Intelligence
Mobile PWA offline mode
SOC2 Type II certification path (see
[docs/SOC2_READINESS.md](#))
Multi-region deployment
External LoRA fine-tuning pipeline from
JSONL exports

Appendix A - Glossary

Term | Definition

RAG | Retrieval-Augmented Generation

FAISS | Facebook AI Similarity Search (vector index)

BNS | Bharatiya Nyaya Sanhita (replaces IPC)

BNSS | Bharatiya Nagarik Suraksha Sanhita (replaces CrPC)

Open Law | Web-grounded legal research mode

Matter scope | AI limited to one case's documents

Appendix B - Related Documents

docs/PRD.md - Original PRD user stories

docs/blueprint/project-blueprint.md -
Engineering blueprint

REPORT.md - System status report

DEPLOY.md - Deployment guide

SECURITY.md - Security policy

docs/SOC2_READINESS.md - Compliance
roadmap

Appendix C - Document Control

Field | Value

Title | LegalEase Product Design & Technical
Architecture Suite

Version | 1.0

Author | LegalEase Engineering

Status | Approved for investor and client
demonstrations

Appendix D - Expanded Module Specifications

D.1 AI Assistant Module (Complete Specification)

Purpose

The AI Assistant is the primary daily interface for attorneys. It supports four distinct reasoning modes, each with separate LLM backends, retrieval strategies, and output formats.

Modes

Mode | Alias | Backend | Retrieval | Output

Knowledge Base | kb, knowledge_base | Ollama | FAISS
user/matter index | Citations required

Open Law | web, web_search, open_law | Gemini +
fallbacks | Live web search | web_sources[]

Hybrid | hybrid | Ollama + Gemini | FAISS + web |
Merged citations

Deep Case | deep_case | Hybrid + extended | Extended
retrieval | Long-form report

Request Lifecycle (Step by Step)

1. **Authentication** - JWT validated; suspended users rejected (403)
2. **Plan gating** - hybrid/deep_case require Pro or Legal Pro membership
3. **Matter scoping** - validate_chat_scope() ensures user owns matter
4. **Session resolution** - conversation_memory loads thread history
5. **Prompt sanitization** - ai_trust strips injection patterns
6. **Intent parsing** - query_parser extracts section numbers, act names
7. **Mode routing** - mode_router selects execution path
8. **Execution** - KB/web/hybrid pipeline runs

- 9. **Persistence** - chat_history + adaptive_interactions recorded
- 10. **Response** - JSON or SSE stream with interaction_id for feedback

Streaming Protocol (SSE)

The frontend (useChat.ts) opens POST /api/v1/chat/stream and parses events:

meta: thread_id, chat_id, interaction_id
token: incremental text chunks
done: final metadata

Timeout: 180 seconds aligned with nginx proxy_timeout.

Follow-Up Suggestions

Generated from content analysis when answer is substantive. Examples: "Explain in simple language", "Summarize key points", "Show official sources".

Similar Cases

Retrieved from KB chunk metadata when statute/case references detected in indexed documents.

D.2 Matter Workspace Module

Matter Data Model (Full Fields)

Field | Type | Description

matter_id | UUID | Primary key
user_id | TEXT | Owner
org_id | TEXT | Optional firm org
matter_name | TEXT | Display name
case_number | TEXT | Court case number
practice_area | TEXT | Criminal, Civil, Corporate, etc.
status_tier | TEXT | ACTIVE, CLOSED, ARCHIVED
client_name | TEXT | Client display name
opposing_party | TEXT | Opposing side
venue | TEXT | Court / jurisdiction

police_station | TEXT | FIR context
fir_number | TEXT | First Information Report
filing_date | TEXT | ISO date
next_hearing_date | TEXT | ISO date
priority | TEXT | High, Medium, Low
description | TEXT | Free text summary

Sub-Modules per Matter

Each matter exposes a workspace navigation (MatterWorkspaceNav.tsx):

1. **Overview** - Dashboard cards, recent activity
2. **Documents** - Linked uploads, matter-scoped indexing
3. **Timeline** - Manual + AI-generated events
4. **Hearings** - Cause list imports, voice entry
5. **Tasks** - Assignable to team members
6. **Evidence** - Matter evidence records
7. **Entities** - Parties, witnesses, organizations
8. **Contradictions** - AI-detected conflicts
9. **Knowledge** - Matter-scoped KB entries
10. **AI** - Matter-scoped chat shortcut
11. **Discussion** - Matter-linked Firm Chat room
12. **History** - Chat thread history filter
13. **Settings** - Matter members, archive

Matter Intelligence Pipeline

POST /matters/{id}/intelligence/run
triggers:

Entity extraction across linked documents
Timeline suggestion generation
Contradiction scanning
Status tracked in matter_intel_status

D.3 Drafting Studio Module (v2/v3/v4)

Document Lifecycle States

draft --> in_review --> partner_review --> approved --> filed --> archived

Transitions enforced by POST

/drafting/workspace/documents/{id}/status

and v4 transition endpoint.

Version Control

Every content save creates

workspace_draft_versions row:

version_number (monotonic)

content_html snapshot

author user_id

created_at timestamp

Compare via /compare and /compare-v3
endpoints

AI Copilot Commands (v3)

POST

/drafting/workspace/documents/{id}/copilot

accepts natural language commands:

"Add limitation period paragraph"

"Insert standard indemnity clause"

"Convert to formal court format"

Court Bundle (v4)

POST /workspace/v4/court-package

assembles:

Main pleading PDF

Annexures with auto-index

Cover page with matter metadata

Table of contents insertion

Filing Readiness Score

GET /workspace/documents/{id}/filing-readiness returns checklist:

- Required fields populated
 - Signature blocks present
 - Annexure references valid
 - Court-specific formatting warnings
-

D.4 Intake Desk (CRM) Module

Pipeline Stages

Default stages: new_lead, contacted, qualified, proposal, converted, rejected, archived

Kanban board at /intake/board with drag-and-drop stage updates via PATCH /crm/{id}/stage.

Lead Scoring

AI analysis (POST /crm/{id}/analyze) produces:

- lead_score (0-100)
- lead_score_band (hot/warm/cold)
- case_strength assessment
- practice_area classification
- suggested next actions

Conversion to Matter

POST /crm/{id}/convert creates matter with pre-filled:

- client_name from lead
- practice_area from classification
- linked documents from crm_lead_documents
- initial timeline event "Lead converted"

Public Intake

POST /practice/public-intake (optional X-Intake-Key header) captures website

leads into crm_leads with org_id scoping.

D.5 Billing Module

Time Entry Flow

1. POST /billing/entries - log hours with matter_id, narrative, rate
2. AI narrative preview - POST /billing/narrative/preview polishes description
3. Entries appear in billing workspace
4. Select entries for invoice - GET /billing/invoices/prefill
5. POST /billing/invoices/draft - save draft
6. POST /billing/invoices/{id}/finalize - lock invoice
7. GET /billing/invoices/{id}/pdf - download PDF

Trust Accounting

Separate from operating billing:

GET /trust/account?matter_id= - trust balance

POST /trust/transactions - deposit/disbursement with narrative

Trust ledger complies with client fund separation principle

Stripe Integration

GET /subscriptions/plans - public plan list

POST /subscriptions/subscribe - checkout session

POST /billing/stripe/webhook - plan sync

Plans: Free, Pro, Legal Pro with feature gates in plan_enforcement.py

D.6 Litigation Desk Module

Court Day Mission Control

GET /practice/court-day/mission-control
aggregates:

- Today's hearings across all matters
- Cause list matches
- Prep pack links
- Limitation warnings

Cause List Import

POST /practice/court-day/import parses
pasted cause list text or uploaded file:

- Extracts case numbers, item numbers,
bench, dates
- Matches to existing matters by
case_number fuzzy match
- Creates matter_hearings records

eCourts Integration

GET /practice/ecourts/search - search by
CNR or party name

GET /practice/ecourts/case/{cnr} - case
preview

POST /practice/ecourts/case/{cnr}/sync -
sync to matter

War Room

GET /practice/litigation/war-
room/{matter_id} - consolidated view:

- Upcoming hearings
- Recent orders
- Open tasks
- Key evidence items
- AI litigation assist (POST
/practice/litigation/ai)

D.7 Enterprise Workspace Module

DMS (Document Management)

Folder hierarchy: ent_dms_folders with parent_id tree

Documents: ent_dms_documents with OCR text, tags, matter_id link

Versions: ent_dms_versions for check-in/check-out pattern

Search: GET
/enterprise/workspace/documents/search
full-text

Court Orders Repository

Upload: POST /enterprise/workspace/court-orders (multipart)

Fields: case_number, court, judge, order_date, summary, content_text

AI analysis: intelligence_json with extracted ratio, directions, next dates

Search: GET /enterprise/workspace/court-orders/search

Knowledge Base

Entry types: precedent, template, memo, statute_note, court_order

Linked to matters and court orders

Powers Evidence Intelligence court-order matcher

Client Portal Operations

POST /enterprise/workspace/client-portal/document-request - ask client for doc

POST /enterprise/workspace/client-portal/request-review - draft approval workflow

Tracked in ent_client_requests, ent_client_approvals

D.8 Firm Chat (Collaboration) Module

Room Types

****Channel**** - firm-wide or practice-area

****DM**** - direct message between two users

****Matter room**** - auto-linked to
matter_id

Real-Time Transport

SSE: GET /collaboration/rooms/{id}/stream
for message stream

WebSocket: /collaboration/ws for typing,
presence (access_token query param)

Fallback polling if SSE unavailable

Permissions (collab_rbac.py)

All authenticated org members can
view/post by default.

DM requires accepted collab_chat_request.

Message Attachments

POST /collaboration/rooms/{id}/messages/{id}/attachments

Stored with download via GET
/collaboration/attachments/{id}/download

AI Summarize

POST /collaboration/rooms/{id}/summarize -
LLM summary of recent messages for catch-up

D.9 Evidence Intelligence Center (Full Specification)

Supported File Formats

PDF, DOCX, DOC, PNG, JPG, JPEG, WEBP, GIF,
TIFF, XLSX, XLS, CSV, TXT, EML, MSG, ZIP

Processing Pipeline (evidence_extraction.py + evidence_intelligence.py)

****Stage 1 - Extraction****

PDF: PyMuPDF native + sparse OCR via pdf_extraction.py

Images: EasyOCR via ocr_engine.py

DOCX: python-docx paragraphs

EML: email.parser headers + body

MSG: extract_msg if installed, else fallback

XLSX: openpyxl or XML fallback

ZIP: recursive extraction (depth limit 2)

****Stage 2 - Metadata****

SHA-256 hash (chain of custody)

Author, created/modified dates

File type, page count

Extraction method tag

****Stage 3 - Classification Categories****

FINANCIAL, CONTRACT, COMMUNICATION, INVOICE, COURT_ORDER, MEDICAL, IDENTITY, EVIDENCE, GENERAL

****Stage 4 - Risk Detection****

FRAUD, BRIBERY, MONEY_TRAIL, CONSPIRACY, HARASSMENT, THREAT, CONTRACT_BREACH

****Stage 5 - Privilege Detection****

ATTORNEY_CLIENT, WORK_PRODUCT, CONFIDENTIAL markers

****Stage 6 - Entity Extraction****

People (Mr./Dr./Shri patterns), Organizations (Pvt Ltd, LLP), Locations (Indian cities), Emails, Phones, IFSC, Bank accounts, Case numbers (WP, CRL, etc.), Dates

****Stage 7 - Timeline****

Date regex extraction from sentences, sorted chronologically

****Stage 8 - Statute Mapping****

Pattern rules mapping content to BNS 316, 318, 336, 61, ICA s.73, BNSS bail

provisions

****Stage 9 - Storage****

discovery_items extended columns:

metadata_json, entities_json,
timeline_json, statutes_json,
privilege_json, risks_json, category,
file_hash

****Stage 10 - Court Order Match****

search_court_orders() + search_knowledge()
from enterprise_workspace

Evidence Strength Score

Combines triage_document() relevance with
risk boost:

80%+ = Highly Relevant

60-79% = Moderately Relevant

40-59% = Low Relevance

Below 40% = Minimal Relevance

Contradiction Detector Algorithm

Compares two documents for:

Awareness conflicts (denial vs
affirmation)

Date conflicts across documents

Amount conflicts (INR values)

Witness statement conflicts (same person,
different claims)

API Endpoints (Complete)

Method | Path | Body | Response

GET | /ediscovery/evidence/formats | - | {formats:[]}

POST | /ediscovery/evidence/upload | multipart
file+matter_id | {analysis, batch_id, items}

POST | /ediscovery/evidence/analyze | {text,
matter_id} | {analysis, court_orders}

GET | /ediscovery/evidence/repository | ?matter_id= |
{items, timeline, count}

GET | /ediscovery/evidence/timeline | ?matter_id= |
{timeline:[]}

POST | /ediscovery/evidence/contradictions |
{document_a, document_b} | {contradictions, summary}

POST | /ediscovery/evidence/statutes | {text} |
{statutes, count}

D.10 Adaptive Learning System

Tables

adaptive_interactions - every chat turn

adaptive_feedback - explicit signals
(thumbs, export, copy)

adaptive_chunk_boosts - per-chunk score
adjustments

adaptive_query_patterns - successful
query expansions

adaptive_mode_stats - per-mode hit rates

learning_signal_events - unified signal
log

human_labels - RLAIF training labels

preference_pairs - DPO training pairs

Signal Types

Positive: thumbs_up, helpful,
verbal_positive, copy, export_docx,
export_pdf, save_to_matter

Negative: thumbs_down, verbal_negative,
regenerate

Background Processing

FEEDBACK_FAST=1 - instant SQLite/Postgres
write, heavy enrichment in daemon threads

FEEDBACK_SKIP_RLAIF=1 - skip Gemini RLAIF
on fast path

Coach pipeline (gemini_ollama_coach.py)
runs offline - never at inference time.

Export for Fine-Tuning

POST /learning/tuning/export - JSONL pairs

POST /learning/training/export-sft - SFT
dataset

POST /learning/training/export-dpo - DPO
preference dataset

Appendix E - Complete API Route Index by Module

E.1 Authentication & Account (main.py + account.py)

POST /auth/login, POST /auth/register, GET /auth/me

POST /account/forgot-password, POST /account/reset-password/{token}

GET /account/export (GDPR ZIP), DELETE /account

GET /account/onboarding, POST /account/onboarding/dismiss

PATCH /account/preferences/learner-mode

E.2 Health & Diagnostics

GET /health/live, /health/ready, /health/llm, /health/embeddings, /health/public

GET /health (user-scoped), GET /health/diagnostics, GET /health/security

GET /health/schema (superadmin), GET /metrics (metrics access)

E.3 Chat & Sessions

POST /chat, POST /chat/stream, POST /chat/export-report

GET /sessions/history, DELETE /sessions/history

GET /sessions/threads/{id}, DELETE /sessions/threads/{id}

POST /sessions/threads/{id}/attachment

E.4 Documents & KB

POST /documents/upload, GET /documents, DELETE /documents/{id}

POST /documents/kb/reindex-auto, GET

/documents/kb/health

GET /documents/jobs/{id}, GET /kb/debug-query

E.5 Matters (53 routes)

Full CRUD plus timeline, hearings, tasks, deadlines, evidence, entities, contradictions, intelligence pipeline, export, audit, members, hearing prep pack, client status letter, cause list import, voice hearing entry.

E.6 Drafting (80+ routes across v2/v3/v4)

Workspace CRUD, versions, export, review, comments, AI generate/assist, autofill, insights, clause-intel, copilot, redline, precedents, court bundle, signatures, track changes, partner review, filing readiness.

E.7 CRM (29 routes)

Dashboard, kanban, analytics, lead CRUD, stage updates, analyze, convert, documents, interactions, follow-up templates, intent correction.

E.8 Billing & Subscriptions

Time entries, expenses, invoices, PDF, Stripe checkout, portal, plans, trust ledger.

E.9 Collaboration (32 HTTP + WS)

Rooms, messages, reactions, attachments, DM requests, presence, typing, search, notifications, summarize.

E.10 Enterprise (23 + 12 routes)

Branding, court sync, agents, pilot program, workspace dashboard, DMS, court orders, knowledge, client portal, audit, storage.

E.11 Practice / Litigation (48 routes)

Court day, cause lists, eCourts, evidence desk, limitation, litigation dashboard, calendar, war room, watchlist, public intake.

E.12 Learning (42 routes)

Feedback, signals, corrections, tuning export, neural training, coach, automation, quality gate, preferences, training pipelines.

E.13 Ediscovery (18 routes)

Evidence upload/analyze/repository/timeline/contradictions/statutes/court-orders, batches, PII tools.

E.14 IPC-BNS v3 (12 routes)

Search, compare, convert, bulk, document upload, report export, categories, matter migration.

E.15 Speech

GET /speech/status, POST /speech/transcribe, POST /speech/polish

E.16 Admin (superadmin)

Users list, suspend/unsuspend, plan override, audit, usage, health.

Appendix F - Database Column Reference (Key Tables)

F.1 users

id TEXT PK, username UNIQUE, password_hash
BYTEA/TEXT, membership TEXT, role TEXT,
email, display_name, suspended INTEGER,
accepted_terms_at, created_at,
last_login_at

F.2 matters

matter_id TEXT PK, user_id TEXT, org_id
TEXT, matter_name TEXT, case_number TEXT,
practice_area TEXT, status_tier TEXT,
client_name TEXT, opposing_party TEXT,
venue TEXT, created_at TEXT, updated_at
TEXT, plus criminal/civil extended fields
(fir_number, police_station,
next_hearing_date, priority, description,
is_archived)

F.3 documents

doc_id TEXT PK, user_id TEXT, title TEXT,
filename TEXT, doc_type TEXT, matter_id
TEXT, content_hash TEXT, indexed INTEGER,
chunk_count INTEGER, created_at TEXT

F.4 discovery_items (Evidence Intelligence)

item_id TEXT PK, batch_id TEXT FK,
source_identifier TEXT, content_payload
TEXT, assigned_tags TEXT, relevance_score
REAL, classification TEXT, rationale TEXT,
reviewed_status INTEGER, created_at TEXT,
file_type TEXT, file_hash TEXT,
metadata_json TEXT, entities_json TEXT,
timeline_json TEXT, statutes_json TEXT,
privilege_json TEXT, risks_json TEXT,
category TEXT, extraction_method TEXT

F.5 workspace_drafts

draft_id TEXT PK, user_id TEXT, matter_id TEXT, title TEXT, document_type TEXT, status TEXT, content_html TEXT, created_at TEXT, updated_at TEXT

F.6 crm_leads

lead_id TEXT PK, user_id TEXT, org_id TEXT, client_name TEXT, email TEXT, phone TEXT, practice_area TEXT, pipeline_stage TEXT, lead_score INTEGER, analysis_json TEXT, matter_id TEXT (after conversion), created_at TEXT

F.7 chat_history

id TEXT PK, user_id TEXT, question TEXT, answer TEXT, language TEXT, mode TEXT, thread_id TEXT, matter_id TEXT, created_at TEXT

F.8 adaptive_interactions

id TEXT PK, user_id TEXT, mode TEXT, query TEXT, query_norm TEXT, answer_preview TEXT, found_in_kb INTEGER, chunk_keys TEXT, chat_id TEXT, thread_id TEXT, scope_key TEXT, created_at TEXT

Appendix G - User Interface Map

G.1 Navigation Structure (Sidebar)

Label | Route | Learner Hidden

AI Assistant | / | No
Dashboard | /dashboard | No
Documents | /documents | No
Matters | /matters | No
Litigation | /litigation | Yes
Intake Desk | /intake | Yes
Firm Chat | /collaboration | Yes
Billing | /billing | Yes
Evidence | /discovery | Yes
Drafting | /drafting | Yes
Legal Tools | /tools | No
Enterprise | /enterprise | Yes
Analytics | /analytics | Yes
Settings | /settings | No
Admin | /admin | Yes (role gated)

G.2 Settings Sub-Pages

Profile and preferences
Memory management (MemoryPanel.tsx) -
persona, facts, pruning
Subscription (/settings/subscription) -
Stripe plans
Team (/settings/team) - org invites

G.3 Design System

Reference: docs/DESIGN_SYSTEM.md
Primary brand: navy (#1e3a5f)
Typography: system fonts, monospace for
legal citations
Components: PageHeader, VoiceTextarea,
ThumbsFeedback, MatterWorkspaceNav

Appendix H - Interview Preparation (System Design Q&A)

H.1 "Walk me through what happens when a user asks a legal question"

The user submits a POST to /api/v1/chat with mode=knowledge_base. FastAPI validates JWT, checks plan, resolves matter scope. chat_service loads thread history from conversation_memory. mode_router confirms KB mode. kb_pipeline runs intent detection (section lookup vs comparison vs summary). kb_retrieval_coordinator retrieves top-k chunks from FAISS using hybrid dense+sparse+MMR. Chunks are reranked, filtered by confidence threshold. Ollama generates answer with citation instructions. Response validated for substantive content. Turn saved to chat_history and adaptive_interactions. interaction_id returned for thumbs feedback.

H.2 "How do you prevent hallucination in legal answers?"

Four layers: (1) RAG grounding - LLM only sees retrieved chunks, (2) NOT_FOUND path when confidence below threshold, (3) kb_gemini_safety blocks cloud LLM from KB synthesis, (4) export quality gate and citation requirements in prompts.

H.3 "How does multi-tenancy work?"

Every database query includes user_id from JWT. Matters, documents, FAISS indexes are namespaced by user_id. Org membership adds org_id scoping for firm features. Matter policy enforces write access by org role. No cross-user vector index access.

H.4 "Why not fine-tune the LLM on every feedback click?"

Latency and cost. Real-time GPU fine-tuning is impractical for SaaS. Instead, chunk boosts and query expansions adjust retrieval immediately. Optional offline JSONL export enables batch LoRA training without affecting production inference.

H.5 "How would you scale to 10,000 users?"

Horizontal API workers behind nginx load balancer, PostgreSQL read replicas, Redis cluster for sessions, dedicated ML worker pool for indexing, S3 for document storage, separate FAISS service or migrate to Pinecone/Weaviate, CDN for frontend static assets.

H.6 "Explain the Evidence Intelligence vs Relativity difference"

Relativity processes terabytes with distributed OCR clusters. LegalEase targets SMB Indian firms with integrated matter+AI context. Our pipeline runs on single EC2 with rule-based+OCR classification, entity extraction, BNS statute mapping, and matter linkage - not just review tags but legal intelligence specific to Indian practice.

Appendix I - Operational Runbooks (Summary)

I.1 Production Deploy

1. Run `scripts/aws_update.ps1 -PublicUri "https://legalease.duckdns.org"`
2. Wait for Docker rebuild (5-15 min)
3. Verify `GET /api/v1/health/live`
4. Hard refresh browser

I.2 Laptop Setup

1. Copy `.env.local.example` to `.env.local`
2. Run `scripts/setup_local_env.ps1`
3. `run_backend.ps1 + run_web.ps1`
4. Open `http://localhost:3000`

I.3 KB Reindex

`POST /documents/kb/reindex-auto` or trigger via Documents UI. Monitor `GET /documents/jobs/{id}`.

I.4 Feedback Not Working (Postgres)

Ensure `adaptive_mode_stats` has `not_found_count` and `threshold_delta` columns. Ensure `adaptive_interactions` insert commits before stats bump. Check `SAAS_USE_POSTGRES_LEGACY=1`.

I.5 OCR Failures

Verify `OCR_ENABLED=1`, EasyOCR installed, `ffmpeg` present in Docker image. Check `GET /health/gpu` for STT/OCR status.

Reference: `docs/runbooks/production-dry-run-checklist.md`, `RUNBOOK.md`, `docs/SUPPORT_RUNBOOK.md`

Appendix J - Compliance & Risk

J.1 SOC2 Readiness

See docs/SOC2_READINESS.md for control mapping:

Access control (JWT, RBAC, admin suspension)

Audit logging (matter_audit_log, ent_audit, audit_events)

Encryption in transit (HTTPS/TLS)

Data export (GDPR GET /account/export)

Incident response (RUNBOOK.md)

J.2 Risk Register Summary

See docs/RISK_REGISTER.md for:

LLM hallucination risk - mitigated by RAG + NOT_FOUND

Data breach - mitigated by tenant isolation + HTTPS

Dependency on Gemini API - mitigated by provider chain fallbacks

Single EC2 point of failure - mitigated by backup/runbook

J.3 Data Retention

User-controlled via account deletion (DELETE /account).

Matters support archive (is_archived flag).

Chat history deletable per thread.

Appendix K - Future Roadmap (Detailed)

K.1 Q3 2026 - Enterprise Hardening

- SSO mandatory for enterprise tier
- SCIM user provisioning (stub exists at /scim/v2)
- Dedicated VPC deployment option
- Automated Postgres backups to S3

K.2 Q4 2026 - AI Advancement

- LLM privilege review in Evidence Intelligence
- Multi-document cross-reference graph
- Voice-first mobile PWA
- Hindi/regional language STT and answers

K.3 2027 - Platform Expansion

- API marketplace for third-party integrations
- White-label firm branding (partial - org branding exists)
- Court e-filing connector (where APIs available)
- External expert network marketplace

Appendix L - Glossary (Extended)

Term | Definition

Cause list | Daily court schedule listing cases to be heard

CNR | Case Number Record (eCourts identifier)

FIR | First Information Report (criminal)

RAG | Retrieval-Augmented Generation - LLM + document search

FAISS | Vector similarity index library

SSE | Server-Sent Events for streaming

RLAIF | Reinforcement Learning from AI Feedback

DPO | Direct Preference Optimization (training method)

SFT | Supervised Fine-Tuning

DMS | Document Management System

BNSS | Bharatiya Nagarik Suraksha Sanhita

BNS | Bharatiya Nyaya Sanhita

IPC | Indian Penal Code (legacy)

CrPC | Code of Criminal Procedure (legacy)

JWT | JSON Web Token authentication

OCR | Optical Character Recognition

MMR | Maximal Marginal Relevance (diversity in retrieval)

KB | Knowledge Base mode in chat

Open Law | Web research mode using live search

Matter scope | Restricting AI to one case's documents

Evidence Strength | Relevance score 0-100% in Evidence Intelligence

Privilege review | Checking if document is attorney-client protected

War room | Litigation command center view for a matter

Pilot program | Enterprise onboarding program (superadmin managed)

Appendix M - Environment Variables Reference

M.1 Core Application

Variable | Default | Purpose

DATABASE_URL | sqlite | PostgreSQL connection string

LEGALEASE_DB_PATH | legalease.db | SQLite file path

SAAS_USE_POSTGRES_LEGACY | auto | Route legacy tables to Postgres

SAAS_PRODUCTION | 0 | Production mode guards

LEGALEASE_LOCAL_DEV | 1 | Laptop dev bypass

REDIS_URL | - | Session store + job queues

JWT_SECRET | - | JWT signing key (min 32 chars prod)

LEGALEASE_TOKEN_TTL | 86400 | Token lifetime seconds

CORS_ORIGINS | localhost | Allowed frontend origins

PUBLIC_APP_URL | - | Public frontend URL

NEXT_PUBLIC_API_URL | - | Frontend API base

M.2 LLM & AI

Variable | Laptop | EC2

LLM_BACKEND | ollama | gemini

OLLAMA_BASE_URL | http://127.0.0.1:11434 | -

OLLAMA_MODEL | legalease-tuned | -

OLLAMA_AUTO_START | 1 | 0

OLLAMA_NUM_GPU | -1 (all) | 0

GEMINI_API_KEY | optional | required

GEMINI_FREE_MODEL | gemini-2.0-flash | same

CLOUD_GEMINI_KB | 0 | 1

GEMINI_KB_SYNTHESIS | 0 | 0

TAVILY_API_KEY | optional | optional

OPENROUTER_API_KEY | optional | optional

M.3 RAG & Retrieval

Variable | Default | Purpose

HF_EMBEDDING_MODEL | all-MiniLM-L6-v2 | Embedding model

RAG_CHUNK_SIZE | 800 | Chunk size chars

RAG_CHUNK_OVERLAP | 120 | Overlap chars

RAG_RETRIEVAL_K | 12 | Initial retrieval count

RAG_FINAL_TOP_K | 6 | Final chunks to LLM

RAG_SCORE_THRESHOLD | 0.35 | Min similarity score

RAG_ENABLE_CROSS_ENCODER | 0 | Reranker (slow on CPU)
KB_CACHE_TTL_SEC | 300 | Query cache TTL
OCR_ENABLED | 1 | Enable OCR pipeline

M.4 Security & Rate Limits

Variable | Default | Purpose

RATE_LIMIT_ENABLED | 1 | Global rate limit
RATE_LIMIT_PER_MINUTE | 120 | General API limit
RATE_LIMIT_CHAT_PER_MINUTE | 30 | Chat-specific limit
SECURITY_HEADERS_ENABLED | 1 | Security headers
FORCE_HTTPS | 0 | HSTS in dev
FIREWALL_ENABLED | 0 | IP allowlist
SUPERADMIN_USERNAMES | - | Admin allowlist
SAAS_ALL_FEATURES_FREE | 0 | Bypass plan gates (EC2 demo)

M.5 Billing & Email

Variable | Purpose

STRIPE_SECRET_KEY | Stripe API
STRIPE_WEBHOOK_SECRET | Webhook verification
STRIPE_PRICE_PRO | Pro plan price ID
STRIPE_PRICE_LEGAL_PRO | Legal Pro price ID
ALLOW MOCK BILLING | Dev mock billing
BREVO_API_KEY | Transactional email
SMTP_HOST | Email fallback

Appendix N - Sample API Request/Response Library

N.1 Register User

Request:

POST /api/v1/auth/register

```
{"username":"advocate1","password":"Secure  
Pass123!","confirm_password":"SecurePass12  
3!","accept_terms":true,"email":"advocate1  
@firm.com"}
```

Response 200:

```
{"ok":true,"user_id":"uuid","token":"eyJ..  
."}
```

N.2 Create Matter

Request:

POST /api/v1/matters

```
{"matter_name":"State v. Sharma","practice  
_area":"Criminal","client_name":"Rajesh  
Sharma","venue":"Sessions Court Delhi"}
```

Response 200:

```
{"matter_id":"uuid","matter_name":"State  
v. Sharma",...}
```

N.3 Upload Document

Request:

POST /api/v1/documents/upload (multipart)

file: contract.pdf

matter_id: uuid (optional)

Response 200:

```
{"doc_id":"uuid","indexed":true,"chunk_cou  
nt":42,"job_id":"optional"}
```

N.4 Knowledge Base Chat

Request:

POST /api/v1/chat

```
{"message": "What does Section 302 IPC say?",  
  "mode": "knowledge_base", "lang": "English",  
  "history": []}
```

Response 200:

```
{"content": "Section 302 IPC deals with...",  
  "similar_cases": [], "follow_ups": ["Explain  
in simple language"], "interaction_id": "uui  
d", "thread_id": "uuid"}
```

N.5 Open Law Chat

Request:

POST /api/v1/chat

```
{"message": "Latest Supreme Court judgment  
on bail in economic offences", "mode": "web_  
search", "lang": "English"}
```

Response 200:

```
{"content": "...", "web_sources": [{"title": "  
...", "url": "..."}], "interaction_id": "uuid"  
}
```

N.6 Thumbs Feedback

Request:

POST /api/v1/learning/feedback

```
{"signal": "thumbs_up", "interaction_id": "uu  
id-from-chat"}
```

Response 200:

```
{"ok": true, "feedback_id": "uuid", "queued": t  
rue}
```

N.7 Evidence Upload

Request:

POST /api/v1/ediscovery/evidence/upload
(multipart)

file: vendor_invoice.pdf

matter_id: uuid

Response 200:

```
{"batch_id":"uuid","analysis":{"evidence_strength":{"percent":92,"label":"Highly Relevant"},"entities":{"people":["Rajesh Sharma"],"dates":["17 Jan 2025"]},"privilege":{"privileged":false}},"items":[...]}
```

N.8 IPC-BNS Search

Request:

GET /api/v1/ipc-bns/v3/search?q=cheating

Response 200:

```
{"results":[{"ipc_section":"420","bns_section":"318","offence":"Cheating"}]}
```

N.9 CRM Create Lead

Request:

POST /api/v1/crm

```
{"client_name":"Priya Verma","phone":"9876543210","practice_area":"Family","referral_source":"Website"}
```

Response 200:

```
{"lead_id":"uuid","pipeline_stage":"new_lead"}
```

N.10 Billing Time Entry

Request:

POST /api/v1/billing/entries

```
{"matter_id":"uuid","hours":2.5,"narrative":"Drafted bail application","rate":5000}
```


Response 200:

```
{"record_id":"uuid","amount_inr":12500}
```

N.11 Drafting Create Document

Request:

POST /api/v1/drafting/workspace/documents

```
{"title":"Bail Application","document_type":  
"bail_application","matter_id":"uuid"}
```

Response 200:

```
{"draft_id":"uuid","status":"draft"}
```

N.12 Enterprise Court Order Upload

Request:

POST /api/v1/enterprise/workspace/court-
orders (multipart)

file: order.pdf

case_number: WP(C) 1234/2024

court: Delhi High Court

Response 200:

```
{"order_id":"uuid","summary":"Court  
directed..."}
```

N.13 Collaboration Create Channel

Request:

POST /api/v1/collaboration/rooms/channel

```
{"name":"Criminal  
Team","description":"Criminal practice  
discussions"}
```

Response 200:

```
{"room_id":"uuid"}
```

N.14 Portal Access Link

Request:

POST /api/v1/portal/access

```
{"matter_id":"uuid","client_email":"client  
@email.com","expires_days":30}
```

Response 200:

```
{"token":"secure-  
token","url":"https://app/portal/token"}
```

N.15 Speech Transcription

Request:

POST /api/v1/speech/transcribe (multipart)

audio: recording.webm

language: English

Response 200:

```
{"text":"transcribed  
text","method":"whisper"}
```

Appendix O - Test Coverage Summary

O.1 Test Files (tests/)

Approximately 29 test files covering:

test_phase4_saas_rigorous.py -
E-discovery, CRM, billing

test_saas_extensions.py - SaaS feature
extensions

KB retrieval and RAG golden tests

Memory and user facts tests

Auth and bcrypt Postgres tests

IPC-BNS engine tests

OCR and PDF extraction tests

O.2 CI Pipeline

GitHub Actions on push/PR:

Python pytest suite

Next.js production build (npm run build)

Validates TypeScript compilation

O.3 Manual Smoke Tests

POST /documents/kb/smoke-test - KB
retrieval regression

POST /matters/{id}/smoke - Matter
pipeline smoke

GET /health/ready - Embeddings + FAISS +
LLM readiness

Appendix P - Product Development History

P.1 Phase A - Core KB & Chat (Complete)

Fixed chat persistence, refactored kb_pipeline, hybrid retrieval, sidebar saved chats, 126+ tests, CI pipeline.

P.2 Phase B - Premium Suite (Complete)

Witness simulator, precedent tree, BNS auditor, deal rooms, redline engine, PII redactor.

P.3 Phase C - Adaptive Learning (Complete)

Thumbs feedback, chunk boosts, query patterns, JSONL export, MessageFeedback UI.

P.4 Phase D - Enterprise Persistence (Complete)

SQLAlchemy ORM, OCR router, analytics page, deal room SQL backing.

P.5 Phase E - Memory & Hardening (Complete)

User profiles, facts, thread summaries, past-conversation RAG, prompt budgets, Memory Panel UI.

P.6 Phase F - Frontend Security (Complete)

Next.js 15.5.18, React 19, zero npm audit vulnerabilities, security headers.

P.7 Phase G - Deploy Infrastructure (Complete)

Docker Compose, Redis sessions, PostgreSQL path, nginx TLS, aws_update.ps1.

P.8 Phase H - Practice Operations (Complete)

CRM intake v2, billing invoices, trust ledger, litigation desk, cause lists, eCourts.

P.9 Phase I - Drafting Studio v4 (Complete)

Court bundles, signatures, track changes, partner review, filing readiness, precedents.

P.10 Phase J - Evidence Intelligence Center (Complete)

Replaced text-box e-discovery with professional evidence workspace: OCR upload, classification, entities, timeline, privilege, statutes, contradictions, matter linkage. Deployed EC2 + laptop.

P.11 Current Production State

URL: <https://legalease.duckdns.org>

Stack: Docker (api, web, nginx, postgres, redis) on EC2

Features: All modules free tier enabled (SAAS_ALL_FEATURES_FREE=1)

Feedback: Postgres adaptive_interactions fix deployed

Appendix Q - Investor FAQ

****Q: What is the moat?****

India-specific statute tools + integrated matter-AI-evidence stack + adaptive learning from attorney feedback.

****Q: Who pays?****

Solo lawyers and SMB firms at Pro/Legal Pro SaaS pricing. Enterprise custom for DMS + SSO.

****Q: What is the TAM?****

600,000+ registered advocates in India, majority underserved by Harvey/Relativity pricing.

****Q: GPU costs?****

KB runs on local Ollama (laptop) or CPU; cloud uses Gemini API pay-per-query with daily caps.

****Q: Regulatory risk?****

Positioned as legal information/research tool, not legal advice. Disclaimers on all outputs.

****Q: Data privacy?****

Tenant isolation, HTTPS, GDPR export, optional field encryption, audit logs.

Appendix R - Document Generation Instructions

To regenerate this PDF after editing the markdown source:

```
cd "Legal_AI_Final 3"  
py scripts/generate_architecture_pdf.py
```

Output: docs/LegalEase_Product_Architecture_Suite.pdf

Source: docs/LEGALEASE_PRODUCT_ARCHITECTURE_SUITE.md

To customize output path:

```
py scripts/generate_architecture_pdf.py  
--output  
"C:\path\to\LegalEase_Documentation.pdf"
```

Appendix S - Screen-by-Screen Product Guide

S.1 Login & Registration (/login)

The login page provides username/password authentication with links to registration and forgot-password flows. Registration requires `accept_terms` checkbox for compliance. On success, JWT stored in `localStorage` (`legalease_token`) and user redirected to `/dashboard`. Cinematic background optional from legacy Streamlit migration. Error states: invalid credentials (401), suspended account (403), network unreachable (connection banner from `ApiConnectionProvider`).

S.2 Dashboard (/dashboard)

Central firm overview showing: active matters count, unbilled work amount, CRM pipeline stage distribution, Evidence Intelligence batch count, recent activity feed, quick links to all modules. `PracticeModuleCards` component renders stat cards with deep links. Mobile-responsive grid layout.

S.3 AI Assistant (/)

Primary chat interface with: mode selector (Knowledge Base, Open Law, Hybrid, Deep Case), language selector (English, Hindi, etc.), matter scope dropdown, voice input via `VoiceTextarea`, streaming markdown rendering, source citations panel, similar cases chips, follow-up suggestion buttons, thumbs feedback (`MessageFeedback` component), thread history in sidebar, export report button. SSE streaming via `useChat` hook. Session restored from URL `?thread=uuid` parameter.

S.4 Documents (/documents)

Global document library: upload PDF/image, view indexing status, KB health indicator, reindex button, document list with filename/size/date, delete document, view extracted timeline and entities per document. Links to matter assignment. Shows chunk count after indexing completes.

S.5 Matters List (/matters)

Table/cards of all matters with: matter name, client, practice area, status, next hearing date, priority badge. Create new matter button links to /matters/new. Filter and search. Archive indicator for closed matters.

S.6 Matter Workspace (/matters/[id])

Tabbed navigation (MatterWorkspaceNav): Overview shows dashboard cards, recent documents, upcoming hearings, open tasks. Each sub-route loads matter-scoped data from /api/v1/matters/{id}/... endpoints.

S.7 Matter Documents (/matters/[id]/documents)

Matter-scoped document list, upload to matter, link unlinked documents, reindex matter KB, document metadata edit. Matter FAISS index separate from global KB.

S.8 Matter Timeline (/matters/[id]/timeline)

Chronological event list with manual add form. AI generate timeline button. Suggestion approval workflow for AI-proposed events. Export capability.

S.9 Matter Hearings (/matters/[id]/hearings)

Hearing list with date, court, item number, bench. Import cause list. Voice entry for hearing details. Link to hearing prep pack PDF.

S.10 Matter Tasks (/matters/[id]/tasks)

Task kanban or list with assignee, due date, status. Create from Firm Chat message action.

S.11 Matter Evidence (/matters/[id]/evidence)

Matter evidence records distinct from Evidence Intelligence Center repository but linked. Add evidence, extract from documents.

S.12 Matter Entities (/matters/[id]/entities)

Extracted and manual entities: parties, witnesses, organizations. Entity profiles with relationship map.

S.13 Matter Contradictions (/matters/[id]/contradictions)

AI-detected contradictions across matter documents. Extract button runs analysis pipeline.

S.14 Matter Knowledge (/matters/[id]/knowledge)

Matter-scoped knowledge entries linked to enterprise knowledge base.

S.15 Matter AI (/matters/[id]/ai)

Shortcut to AI Assistant pre-scoped to matter_id.

S.16 Matter Discussion (/matters/[id]/discussion)

Matter-linked Firm Chat room for team collaboration on case.

S.17 Evidence Intelligence Center (/discovery)

Five tabs: Upload Evidence (drag-drop), Evidence Repository, Timeline, Contradiction Check, Statute & Orders. Matter dropdown links all evidence. Analysis panel shows strength score, entities, risks, privilege, statutes, metadata hash.

S.18 Intake Desk (/intake)

Lead list with pipeline stage filters. Quick stats. Link to new lead, board view, analytics, public intake portal config.

S.19 Intake Board (/intake/board)

Kanban columns per pipeline stage. Drag card to update stage via API.

S.20 Intake Lead Detail (/intake/[leadId])

Lead profile, documents, interactions timeline, AI analysis panel, convert to matter button, follow-up templates, reject/archive actions.

S.21 Billing (/billing)

Tabs: time entries, expenses, invoices, collections, matter financials. Log time form with AI narrative polish. Invoice builder with PDF preview. Trust account tab for client funds.

S.22 Litigation Desk (/litigation)

Mission control for hearings, cause list import, calendar view, war room access, limitation deadlines, litigation AI assist, watchlist dashboard.

S.23 Drafting List (/drafting)

All drafts with status badges, matter links, create new draft, filter by document type.

S.24 Draft Editor (/drafting/[draftId])

Rich text/HTML editor, version history sidebar, AI copilot panel, clause intelligence, comments, track changes, export buttons, filing readiness score, collaboration presence indicators.

S.25 Legal Tools Hub (/tools)

Cards linking to IPC-BNS converter, court fee calculator, contract review, case prediction, citation builder, ODR tools.

S.26 IPC-BNS Tool (/tools/ipc-bns)

Search IPC/BNS sections, compare migration, bulk convert list, upload document for auto-conversion, export PDF/DOCX report.

S.27 Enterprise (/enterprise)

Dashboard, global search, matters hub, DMS folders/documents, court orders repository, knowledge base, client portal ops, analytics, storage usage, audit log viewer.

S.28 Firm Chat (/collaboration)

Room list (channels + DMs), message thread, attachments, reactions, typing indicators, user presence, AI summarize, create channel/DM, matter-linked rooms.

S.29 Analytics (/analytics)

Learning stats, retrieval metrics, similar case clusters, judicial analytics from judgments table, tuning export triggers.

S.30 Settings (/settings)

Profile edit, memory panel (persona, facts, prune), LLM test button, API connection status, learner mode toggle.

S.31 Subscription (/settings/subscription)

Current plan, upgrade buttons (Stripe checkout), billing portal link, payment history.

S.32 Team (/settings/team)

Org members list, invite form, pending invites, role management.

S.33 Admin (/admin)

Superadmin: user search, suspend/unsuspend, plan override, system audit, usage metrics, health dashboard.

S.34 Client Portal (/portal/[token])

Public-facing (no login): matter status, timeline, document upload area, e-sign stub, branded header from org branding.

S.35 Onboarding (/onboarding)

First-run checklist: upload first document, create first matter, try AI chat, explore modules. Dismissible.

Appendix T - Error Code Reference

HTTP Code | Meaning | Common Causes

400 | Bad Request | Validation failure, missing matter_id, empty file

401 | Unauthorized | Missing/expired JWT

403 | Forbidden | Suspended account, plan gate, matter access denied

404 | Not Found | Matter/document/batch not found

422 | Unprocessable | Pydantic validation error

429 | Too Many Requests | Rate limit exceeded

500 | Server Error | Unhandled exception, DB error

503 | Service Unavailable | LLM unavailable, not ready

Common Error Messages

"Backend not reachable" - Frontend cannot connect to API (check NEXT_PUBLIC_API_URL)

"interaction not found" - Feedback before interaction persisted (Postgres transaction fix applied)

"Matter not found" - Invalid matter_id or wrong user scope

"Plan required" - Hybrid/deep_case on Free tier

"NOT_FOUND in knowledge base" - No relevant chunks above threshold (expected behavior)

"Gemini daily limit reached" - GEMINI_DAILY_* quota exhausted

Appendix U - Architecture Decision Records (ADR)

ADR-001: Ollama for KB, Gemini for Web

Decision: Separate LLM backends by mode to prevent cloud hallucination in document-grounded answers.

Status: Accepted. Enforced by `kb_gemini_safety.py`.

ADR-002: FAISS over Pinecone for v1

Decision: Local FAISS indexes for zero vendor dependency and laptop offline dev.

Status: Accepted. Migration to managed vector DB in roadmap.

ADR-003: SQLite default, Postgres production

Decision: Zero-config laptop dev; Postgres when `SAAS_USE_POSTGRES_LEGACY=1`.

Status: Accepted. EC2 runs full Postgres.

ADR-004: Feedback via retrieval tuning not live fine-tune

Decision: Chunk boosts and query patterns instead of GPU training per click.

Status: Accepted. JSONL export for optional batch training.

ADR-005: Evidence Intelligence rule-based classification

Decision: Regex/pattern classification for predictable legal tags without GPU on EC2.

Status: Accepted. LLM enrichment planned for v2.

ADR-006: JWT stateless auth with DB membership refresh

Decision: Stateless JWT for horizontal scaling; membership from DB not token for plan changes.

Status: Accepted.

ADR-007: Next.js over Streamlit for production UI

Decision: Streamlit retained for legacy; Next.js 15 is primary product surface.

Status: Accepted.

End of Documentation Suite

Appendix V - Complete API Route Index (Auto-Generated)

Total routes: ****493**** (includes WebSocket endpoints).

Base prefix: ``/api/v1``. Authentication:
JWT Bearer unless noted public.

Module | Method | Path

account | DELETE | ``/api/v1/account``
account | GET | ``/api/v1/account/export``
account | POST | ``/api/v1/account/forgot-password``
account | GET | ``/api/v1/account/onboarding``
account | POST | ``/api/v1/account/onboarding/dismiss``
account | GET | ``/api/v1/account/preferences``
account | PATCH |
``/api/v1/account/preferences/learner-mode``
account | POST | ``/api/v1/account/reset-password/{token}``
account | POST | ``/api/v1/account/verify-email/confirm``
account | POST | ``/api/v1/account/verify-email/send``
admin | GET | ``/api/v1/admin/audit``
admin | GET | ``/api/v1/admin/health``
admin | GET | ``/api/v1/admin/usage``
admin | GET | ``/api/v1/admin/users``
admin | POST | ``/api/v1/admin/users/{user_id}/plan``
admin | POST |
``/api/v1/admin/users/{user_id}/suspend``
admin | POST |
``/api/v1/admin/users/{user_id}/unsuspend``
billing | GET | ``/api/v1/billing/collections``
billing | GET | ``/api/v1/billing/entries``
billing | POST | ``/api/v1/billing/entries``
billing | POST | ``/api/v1/billing/entries/bulk``
billing | DELETE |
``/api/v1/billing/entries/{record_id}``
billing | PUT | ``/api/v1/billing/entries/{record_id}``
billing | POST |
``/api/v1/billing/entries/{record_id}/duplicate``
billing | GET | ``/api/v1/billing/expenses``
billing | POST | ``/api/v1/billing/expenses``
billing | DELETE |
``/api/v1/billing/expenses/{expense_id}``
billing | GET | ``/api/v1/billing/invoices``
billing | POST | ``/api/v1/billing/invoices``
billing | POST | ``/api/v1/billing/invoices/compute-totals``
billing | POST | ``/api/v1/billing/invoices/draft``

billing | GET | `/api/v1/billing/invoices/prefill`
 billing | GET |
 `/api/v1/billing/invoices/{invoice\_id}`
 billing | PUT |
 `/api/v1/billing/invoices/{invoice\_id}`
 billing | POST |
 `/api/v1/billing/invoices/{invoice\_id}/finalize`
 billing | GET |
 `/api/v1/billing/invoices/{invoice\_id}/pdf`
 billing | PATCH |
 `/api/v1/billing/invoices/{invoice\_id}/status`
 billing | GET |
 `/api/v1/billing/matter/{matter\_id}/financials`
 billing | GET |
 `/api/v1/billing/matter/{matter\_id}/profile`
 billing | PUT |
 `/api/v1/billing/matter/{matter\_id}/profile`
 billing | POST | `/api/v1/billing/narrative/correct`
 billing | POST | `/api/v1/billing/narrative/preview`
 billing | GET | `/api/v1/billing/payments`
 billing | GET | `/api/v1/billing/plans`
 billing | GET |
 `/api/v1/billing/reports/{report\_type}`
 billing | GET | `/api/v1/billing/status`
 billing | GET | `/api/v1/billing/summary`
 billing | GET | `/api/v1/billing/workspace`
 chat | POST | `/api/v1/chat`
 chat | POST | `/api/v1/chat/export-report`
 chat | POST | `/api/v1/chat/stream`
 clauses | GET | `/api/v1/clauses`
 clauses | POST | `/api/v1/clauses`
 clauses | POST | `/api/v1/clauses/feedback`
 collab | GET | `/api/v1/collaboration/attachments/{attachment\_id}/download`
 collab | GET | `/api/v1/collaboration/debug/realtime`
 collab | GET | `/api/v1/collaboration/members`
 collab | POST |
 `/api/v1/collaboration/messages/{message\_id}/create-deadline`
 collab | POST |
 `/api/v1/collaboration/messages/{message\_id}/create-task`
 collab | GET | `/api/v1/collaboration/notifications`
 collab | POST | `/api/v1/collaboration/notifications/{notification\_id}/read`
 collab | GET | `/api/v1/collaboration/permissions`
 collab | GET | `/api/v1/collaboration/presence`
 collab | POST | `/api/v1/collaboration/presence`
 collab | GET | `/api/v1/collaboration/requests`
 collab | POST | `/api/v1/collaboration/requests`
 collab | POST |

```

`/api/v1/collaboration/requests/{request_id}/accept`
collab | POST |
`/api/v1/collaboration/requests/{request_id}/reject`
collab | GET | `/api/v1/collaboration/rooms`
collab | POST | `/api/v1/collaboration/rooms/channel`
collab | POST | `/api/v1/collaboration/rooms/dm`
collab | GET |
`/api/v1/collaboration/rooms/matter/{matter_id}`
collab | GET |
`/api/v1/collaboration/rooms/{room_id}`
collab | GET |
`/api/v1/collaboration/rooms/{room_id}/context`
collab | GET |
`/api/v1/collaboration/rooms/{room_id}/messages`
collab | POST |
`/api/v1/collaboration/rooms/{room_id}/messages`
collab | POST | `/api/v1/collaboration/rooms/{room_id}
/messages/{message_id}/attachments`
collab | POST | `/api/v1/collaboration/rooms/{room_id}
/messages/{message_id}/reactions`
collab | POST |
`/api/v1/collaboration/rooms/{room_id}/read`
collab | GET |
`/api/v1/collaboration/rooms/{room_id}/stats`
collab | GET |
`/api/v1/collaboration/rooms/{room_id}/stream`
collab | POST |
`/api/v1/collaboration/rooms/{room_id}/summarize`
collab | GET |
`/api/v1/collaboration/rooms/{room_id}/typing`
collab | GET | `/api/v1/collaboration/search`
collab | POST | `/api/v1/collaboration/typing`
collab | GET | `/api/v1/collaboration/users/search`
collab | WEBSOCKET | `/api/v1/collaboration/ws`
crm | GET | `/api/v1/crm`
crm | POST | `/api/v1/crm`
crm | GET | `/api/v1/crm/analytics`
crm | POST | `/api/v1/crm/assistant`
crm | POST | `/api/v1/crm/classify`
crm | GET | `/api/v1/crm/command-center`
crm | GET | `/api/v1/crm/dashboard`
crm | POST | `/api/v1/crm/intent/correct`
crm | GET | `/api/v1/crm/kanban`
crm | GET | `/api/v1/crm/permissions`
crm | GET | `/api/v1/crm/pipeline-stages`
crm | GET | `/api/v1/crm/{lead_id}`
crm | PATCH | `/api/v1/crm/{lead_id}`
crm | POST | `/api/v1/crm/{lead_id}/analyze`
crm | POST | `/api/v1/crm/{lead_id}/archive`
crm | GET | `/api/v1/crm/{lead_id}/audit`
crm | POST | `/api/v1/crm/{lead_id}/convert`

```

crm | POST | `/api/v1/crm/{lead\_id}/convert-to-matter`
crm | POST | `/api/v1/crm/{lead\_id}/convert/preview`
crm | GET | `/api/v1/crm/{lead\_id}/documents`
crm | POST | `/api/v1/crm/{lead\_id}/documents`
crm | POST | `/api/v1/crm/{lead\_id}/follow-up/apply`
crm | POST | `/api/v1/crm/{lead\_id}/follow-up/preview`
crm | POST | `/api/v1/crm/{lead\_id}/follow-up/send`
crm | GET | `/api/v1/crm/{lead\_id}/follow-up/templates`
crm | GET | `/api/v1/crm/{lead\_id}/interactions`
crm | POST | `/api/v1/crm/{lead\_id}/interactions`
crm | POST | `/api/v1/crm/{lead\_id}/reject`
crm | PATCH | `/api/v1/crm/{lead\_id}/stage`
dashboard | GET | `/api/v1/dashboard/full`
documents | GET | `/api/v1/documents`
documents | POST | `/api/v1/documents/index`
documents | GET | `/api/v1/documents/jobs`
documents | GET | `/api/v1/documents/jobs/{job\_id}`
documents | GET | `/api/v1/documents/kb/health`
documents | POST | `/api/v1/documents/kb/reindex-auto`
documents | POST | `/api/v1/documents/kb/smoke-test`
documents | POST | `/api/v1/documents/kb/sync-status`
documents | POST | `/api/v1/documents/upload`
documents | DELETE | `/api/v1/documents/{doc\_id}`
documents | GET |
`/api/v1/documents/{doc\_id}/entities`
documents | GET |
`/api/v1/documents/{doc\_id}/timeline`
drafting_v3 | POST | `/api/v1/drafting/redline`
drafting_v3 | POST |
`/api/v1/drafting/redline/feedback`
drafting_studio | POST | `/api/v1/drafting/smart-draft/generate`
drafting_studio | GET | `/api/v1/drafting/smart-draft/types`
drafting_studio | GET | `/api/v1/drafting/smart-draft/{draft\_type}/questions`
drafting_workspace | POST |
`/api/v1/drafting/workspace/ai/generate`
drafting_workspace | GET |
`/api/v1/drafting/workspace/dashboard`
drafting_workspace | GET |
`/api/v1/drafting/workspace/document-types`
drafting_workspace | GET |
`/api/v1/drafting/workspace/documents`
drafting_workspace | POST |
`/api/v1/drafting/workspace/documents`
drafting_workspace | DELETE |

``/api/v1/drafting/workspace/documents/{draft_id}``
`drafting_workspace | GET |`
``/api/v1/drafting/workspace/documents/{draft_id}``
`drafting_workspace | PATCH |`
``/api/v1/drafting/workspace/documents/{draft_id}``
`drafting_workspace | POST | `/api/v1/drafting/workspa`
`ce/documents/{draft_id}/ai/assist``
`drafting_v4 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/annexures``
`drafting_v4 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/annexures``
`drafting_v4 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/assign``
`drafting_v4 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/assignments``
`drafting_v4 | PATCH | `/api/v1/drafting/workspace/doc`
`uments/{draft_id}/assignments/{assignment_id}``
`drafting_v3 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/audit``
`drafting_v3 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/autofill``
`drafting_v4 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/billing-session``
`drafting_v3 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/clause-intel``
`drafting_v4 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/collaboration-hub``
`drafting_workspace | POST | `/api/v1/drafting/workspa`
`ce/documents/{draft_id}/comments``
`drafting_v3 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/comments/{comment_id}/resolve``
`drafting_workspace | GET | `/api/v1/drafting/workspac`
`e/documents/{draft_id}/compare``
`drafting_v4 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/compare-precedent``
`drafting_v3 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/compare-v3``
`drafting_v3 | PATCH | `/api/v1/drafting/workspace/doc`
`uments/{draft_id}/content``
`drafting_v3 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/copilot``
`drafting_workspace | POST | `/api/v1/drafting/workspa`
`ce/documents/{draft_id}/export``
`drafting_v3 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/export-v3``
`drafting_v4 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/filing-readiness``
`drafting_v4 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/insert-annexure-index``
`drafting_v4 | POST | `/api/v1/drafting/workspace/docu`
`ments/{draft_id}/insert-toc``
`drafting_v3 | GET | `/api/v1/drafting/workspace/docum`
`ents/{draft_id}/insights``
`drafting_v4 | POST | `/api/v1/drafting/workspace/docu`

ments/{draft_id}/link-hearing`

drafting_v4 | GET | `/api/v1/drafting/workspace/documents/{draft\_id}/links`

drafting_v4 | DELETE | `/api/v1/drafting/workspace/documents/{draft\_id}/lock`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/lock`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/partner-approve`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/partner-review`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/partner-revision`

drafting_v4 | GET | `/api/v1/drafting/workspace/documents/{draft\_id}/presence`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/presence/heartbeat`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/promote-precedent`

drafting_workspace | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/restore/{version\_number}`

drafting_workspace | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/review`

drafting_v4 | GET | `/api/v1/drafting/workspace/documents/{draft\_id}/review-workspace`

drafting_v4 | GET | `/api/v1/drafting/workspace/documents/{draft\_id}/signatures`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/signatures`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/signatures/{signature\_id}/signed`

drafting_v3 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/status`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/suggestions`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/suggestions/{suggestion\_id}/resolve`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/sync-litigation`

drafting_v4 | GET | `/api/v1/drafting/workspace/documents/{draft\_id}/timeline`

drafting_v4 | GET | `/api/v1/drafting/workspace/documents/{draft\_id}/track-changes`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/track-changes`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/track-changes/{change\_id}/resolve`

drafting_v4 | POST | `/api/v1/drafting/workspace/documents/{draft\_id}/transition`

drafting_workspace | GET | `/api/v1/drafting/workspace/documents/{draft\_id}/versions`

drafting_v3 | GET | `/api/v1/drafting/workspace/v3/matters/{matter\_id}/variables`

drafting_v3 | POST |

``/api/v1/drafting/workspace/v3/pack``
 drafting_v3 | GET |
``/api/v1/drafting/workspace/v3/search``
 drafting_v3 | GET |
``/api/v1/drafting/workspace/v3/templates``
 drafting_v3 | POST |
``/api/v1/drafting/workspace/v3/templates/create-document``
 drafting_v3 | POST |
``/api/v1/drafting/workspace/v3/templates/render``
 drafting_v3 | GET |
``/api/v1/drafting/workspace/v3/workflow``
 drafting_v4 | GET |
``/api/v1/drafting/workspace/v4/control-center``
 drafting_v4 | POST |
``/api/v1/drafting/workspace/v4/court-package``
 drafting_v4 | POST | ``/api/v1/drafting/workspace/v4/matters/{matter_id}/court-bundle``
 drafting_v4 | GET | ``/api/v1/drafting/workspace/v4/matters/{matter_id}/drafting``
 drafting_v4 | GET | ``/api/v1/drafting/workspace/v4/matters/{matter_id}/drafting-overview``
 drafting_v4 | POST | ``/api/v1/drafting/workspace/v4/matters/{matter_id}/drafts``
 drafting_v4 | GET |
``/api/v1/drafting/workspace/v4/precedents``
 drafting_v4 | POST |
``/api/v1/drafting/workspace/v4/precedents``
 drafting_v4 | GET |
``/api/v1/drafting/workspace/v4/precedents/search``
 ediscovery | GET | ``/api/v1/ediscovery/batches``
 ediscovery | POST | ``/api/v1/ediscovery/batches``
 ediscovery | GET |
``/api/v1/ediscovery/batches/{batch_id}``
 ediscovery | GET |
``/api/v1/ediscovery/batches/{batch_id}/search``
 ediscovery | POST |
``/api/v1/ediscovery/evidence/analyze``
 ediscovery | POST |
``/api/v1/ediscovery/evidence/contradictions``
 ediscovery | POST |
``/api/v1/ediscovery/evidence/court-orders``
 ediscovery | GET |
``/api/v1/ediscovery/evidence/formats``
 ediscovery | GET |
``/api/v1/ediscovery/evidence/repository``
 ediscovery | POST |
``/api/v1/ediscovery/evidence/statutes``
 ediscovery | GET |
``/api/v1/ediscovery/evidence/timeline``
 ediscovery | POST |
``/api/v1/ediscovery/evidence/upload``
 ediscovery | POST |
``/api/v1/ediscovery/items/{item_id}/review``

ediscovery | GET | `/api/v1/ediscovery/jobs/{job\_id}`
ediscovery | POST | `/api/v1/ediscovery/pii/detect`
ediscovery | POST | `/api/v1/ediscovery/pii/redact`
ediscovery | POST |
`/api/v1/ediscovery/pii/whitelist`
ediscovery | POST | `/api/v1/ediscovery/triage`
engines | GET |
`/api/v1/engines/matters/{matter\_id}/autopilot`
engines | GET | `/api/v1/engines/status`
engines | GET | `/api/v1/engines/watchlist`
engines | POST | `/api/v1/engines/watchlist`
engines | DELETE |
`/api/v1/engines/watchlist/{watch\_id}`
engines | POST |
`/api/v1/engines/watchlist/{watch\_id}/check`
enterprise | GET | `/api/v1/enterprise/agents`
enterprise | POST | `/api/v1/enterprise/agents/run`
enterprise | GET | `/api/v1/enterprise/branding`
enterprise | GET | `/api/v1/enterprise/court/status`
enterprise | POST | `/api/v1/enterprise/court/sync`
enterprise | GET |
`/api/v1/enterprise/orgs/{org\_id}/branding`
enterprise | PATCH |
`/api/v1/enterprise/orgs/{org\_id}/branding`
enterprise | GET | `/api/v1/enterprise/pilot/firms`
enterprise | POST | `/api/v1/enterprise/pilot/firms`
enterprise | PATCH |
`/api/v1/enterprise/pilot/firms/{pilot\_id}`
enterprise | GET | `/api/v1/enterprise/pilot/summary`
enterprise_workspace | GET |
`/api/v1/enterprise/workspace/analytics`
enterprise_workspace | GET |
`/api/v1/enterprise/workspace/audit`
enterprise_workspace | GET |
`/api/v1/enterprise/workspace/client-portal`
enterprise_workspace | POST |
`/api/v1/enterprise/workspace/client-portal/document-
request`
enterprise_workspace | POST |
`/api/v1/enterprise/workspace/client-portal/request-
review`
enterprise_workspace | GET |
`/api/v1/enterprise/workspace/court-orders`
enterprise_workspace | POST |
`/api/v1/enterprise/workspace/court-orders`
enterprise_workspace | GET |
`/api/v1/enterprise/workspace/court-orders/search`
enterprise_workspace | GET |
`/api/v1/enterprise/workspace/court-
orders/{order_id}`
enterprise_workspace | GET |
`/api/v1/enterprise/workspace/dashboard`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/documents`

enterprise_workspace | POST |
`/api/v1/enterprise/workspace/documents`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/documents/search`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/documents/{doc\_id}`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/folders`

enterprise_workspace | POST |
`/api/v1/enterprise/workspace/folders`

enterprise_workspace | POST |
`/api/v1/enterprise/workspace/folders/seed-matter`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/knowledge`

enterprise_workspace | POST |
`/api/v1/enterprise/workspace/knowledge`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/matters`

enterprise_workspace | GET | `api/v1/enterprise/work
space/matters/{matter_id}/hub`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/search`

enterprise_workspace | GET |
`/api/v1/enterprise/workspace/storage`

esign | POST |
`/api/v1/esign/mock/{request\_id}/complete`

esign | POST | `api/v1/esign/requests`

esign | GET | `api/v1/esign/requests/{request\_id}`

feedback | GET | `api/v1/feedback-learning/queue`

feedback | GET | `api/v1/feedback-
learning/queue/mine`

feedback | POST | `api/v1/feedback-
learning/queue/{queue_id}/review`

health | GET | `api/v1/health`

health | GET | `api/v1/health/embeddings`

health | GET | `api/v1/health/gpu`

health | GET | `api/v1/health/live`

health | GET | `api/v1/health/llm`

health | GET | `api/v1/health/public`

health | GET | `api/v1/health/ready`

ipc_bns_v3 | GET | `api/v1/ipc-bns/v3/bns/{section}`

ipc_bns_v3 | POST | `api/v1/ipc-bns/v3/bulk`

ipc_bns_v3 | GET | `api/v1/ipc-bns/v3/categories`

ipc_bns_v3 | GET | `api/v1/ipc-
bns/v3/categories/{category}`

ipc_bns_v3 | GET | `api/v1/ipc-
bns/v3/compare/{ipc_section}`

ipc_bns_v3 | POST | `api/v1/ipc-bns/v3/convert`

ipc_bns_v3 | POST | `api/v1/ipc-
bns/v3/document/convert`

ipc_bns_v3 | POST | `/api/v1/ipc-bns/v3/document/upload`

ipc_bns_v3 | GET | `/api/v1/ipc-bns/v3/ipc/{section}`

ipc_bns_v3 | GET | `/api/v1/ipc-bns/v3/matters/{matter\_id}/migration`

ipc_bns_v3 | GET | `/api/v1/ipc-bns/v3/meta`

ipc_bns_v3 | POST | `/api/v1/ipc-bns/v3/report/export`

ipc_bns_v3 | GET | `/api/v1/ipc-bns/v3/search`

kb_debug | GET | `/api/v1/kb/debug-batch`

kb_debug | GET | `/api/v1/kb/debug-query`

learning | GET | `/api/v1/learning/analytics/full`

learning | GET | `/api/v1/learning/automation/jobs`

learning | GET |
`/api/v1/learning/automation/jobs/{job\_id}`

learning | POST | `/api/v1/learning/automation/run-now`

learning | GET | `/api/v1/learning/automation/status`

learning | POST | `/api/v1/learning/correction`

learning | POST | `/api/v1/learning/engine/auto-improve`

learning | POST | `/api/v1/learning/engine/rescue-test`

learning | GET | `/api/v1/learning/engine/status`

learning | POST | `/api/v1/learning/eval/holdout`

learning | POST | `/api/v1/learning/feedback`

learning | GET | `/api/v1/learning/preferences`

learning | GET | `/api/v1/learning/progress`

learning | GET | `/api/v1/learning/quality-gate`

learning | POST | `/api/v1/learning/signals`

learning | GET | `/api/v1/learning/signals/stats`

learning | GET | `/api/v1/learning/signals/tags`

learning | GET | `/api/v1/learning/stats`

learning | POST | `/api/v1/learning/training/export-dpo`

learning | POST | `/api/v1/learning/training/export-sft`

learning | GET | `/api/v1/learning/training/llm-status`

learning | GET | `/api/v1/learning/training/status`

learning | POST | `/api/v1/learning/training/train-dpo`

learning | POST | `/api/v1/learning/training/train-sft`

learning | POST |
`/api/v1/learning/tuning/coach/analyze`

learning | POST |
`/api/v1/learning/tuning/coach/apply`

learning | GET |
`/api/v1/learning/tuning/coach/directives`

learning | POST |

``/api/v1/learning/tuning/coach/directives``
 learning | POST | ``/api/v1/learning/tuning/coach/run``
 learning | POST |
``/api/v1/learning/tuning/coach/schedule/run-now``
 learning | GET |
``/api/v1/learning/tuning/coach/schedule/status``
 learning | POST |
``/api/v1/learning/tuning/coach/schedule/toggle``
 learning | GET |
``/api/v1/learning/tuning/coach/status``
 learning | POST |
``/api/v1/learning/tuning/coach/toggle``
 learning | POST | ``/api/v1/learning/tuning/export``
 learning | POST | ``/api/v1/learning/tuning/export-saas``
 learning | POST |
``/api/v1/learning/tuning/neural/collect``
 learning | GET |
``/api/v1/learning/tuning/neural/status``
 learning | POST |
``/api/v1/learning/tuning/neural/train``
 learning | POST |
``/api/v1/learning/tuning/ollama/export-modelfile``
 learning | GET |
``/api/v1/learning/tuning/ollama/export-status``
 learning | POST |
``/api/v1/learning/tuning/scope/promote``
 legal_conversion | GET | ``/api/v1/legal-conversion/convert``
 legal_conversion | POST | ``/api/v1/legal-conversion/convert``
 legal_conversion | GET | ``/api/v1/legal-conversion/meta``
 legal_conversion | GET | ``/api/v1/legal-conversion/search``
 matters | GET | ``/api/v1/matters``
 matters | POST | ``/api/v1/matters``
 matters | GET | ``/api/v1/matters/documents/unlinked``
 matters | GET | ``/api/v1/matters/evidence-desk``
 matters | POST | ``/api/v1/matters/evidence-desk/scan``
 matters | GET | ``/api/v1/matters/health/indexing``
 matters | GET | ``/api/v1/matters/hearings/digest``
 matters | GET | ``/api/v1/matters/meta/types``
 matters | GET | ``/api/v1/matters/notifications/all``
 matters | DELETE | ``/api/v1/matters/{matter_id}``
 matters | GET | ``/api/v1/matters/{matter_id}``
 matters | PATCH | ``/api/v1/matters/{matter_id}``
 matters | GET | ``/api/v1/matters/{matter_id}/audit``
 matters | GET |
``/api/v1/matters/{matter_id}/autopilot``
 matters | GET | ``/api/v1/matters/{matter_id}/client-status-letter``

matters | GET |
`/api/v1/matters/{matter\_id}/contradictions`

matters | POST |
`/api/v1/matters/{matter\_id}/contradictions/extract`

matters | GET |
`/api/v1/matters/{matter\_id}/dashboard`

matters | GET |
`/api/v1/matters/{matter\_id}/deadlines`

matters | POST |
`/api/v1/matters/{matter\_id}/deadlines`

matters | POST |
`/api/v1/matters/{matter\_id}/documents/link`

matters | POST |
`/api/v1/matters/{matter\_id}/documents/upload`

matters | PATCH |
`/api/v1/matters/{matter\_id}/documents/{document\_id}`

matters | GET |
`/api/v1/matters/{matter\_id}/entities`

matters | POST |
`/api/v1/matters/{matter\_id}/entities/extract`

matters | GET |
`/api/v1/matters/{matter\_id}/entities/profiles`

matters | GET |
`/api/v1/matters/{matter\_id}/evidence`

matters | POST |
`/api/v1/matters/{matter\_id}/evidence`

matters | POST |
`/api/v1/matters/{matter\_id}/evidence/extract`

matters | GET | ` /api/v1/matters/{matter\_id}/export`

matters | GET | ` /api/v1/matters/{matter_id}/hearing-
prep-pack`

matters | GET |
`/api/v1/matters/{matter\_id}/hearings`

matters | POST |
`/api/v1/matters/{matter\_id}/hearings`

matters | POST |
`/api/v1/matters/{matter\_id}/hearings/extract`

matters | POST |
`/api/v1/matters/{matter\_id}/hearings/from-voice`

matters | POST |
`/api/v1/matters/{matter_id}/hearings/import-cause-
list`

matters | POST |
`/api/v1/matters/{matter\_id}/intelligence/run`

matters | GET |
`/api/v1/matters/{matter\_id}/intelligence/status`

matters | GET | ` /api/v1/matters/{matter\_id}/members`

matters | POST |
`/api/v1/matters/{matter\_id}/members`

matters | POST | ` /api/v1/matters/{matter\_id}/notes`

matters | POST |
`/api/v1/matters/{matter\_id}/restore`

matters | GET | ` /api/v1/matters/{matter\_id}/search`

matters | POST | `/api/v1/matters/{matter\_id}/smoke`
 matters | GET | `/api/v1/matters/{matter\_id}/tasks`
 matters | POST | `/api/v1/matters/{matter\_id}/tasks`
 matters | PATCH |
 `/api/v1/matters/{matter\_id}/tasks/{task\_id}`
 matters | GET |
 `/api/v1/matters/{matter\_id}/timeline`
 matters | POST |
 `/api/v1/matters/{matter\_id}/timeline`
 matters | POST |
 `/api/v1/matters/{matter\_id}/timeline/generate`
 matters | GET |
 `/api/v1/matters/{matter\_id}/timeline/suggestions`
 matters | POST | `/api/v1/matters/{matter_id}/timelin
 e/suggestions/{suggestion_id}/approve`
 matters | POST | `/api/v1/matters/{matter_id}/timelin
 e/suggestions/{suggestion_id}/reject`
 memory | GET | `/api/v1/memory/context`
 memory | POST | `/api/v1/memory/facts`
 memory | POST | `/api/v1/memory/facts/reindex-chats`
 memory | DELETE | `/api/v1/memory/facts/{fact\_id}`
 memory | PATCH | `/api/v1/memory/facts/{fact\_id}`
 memory | GET | `/api/v1/memory/profile`
 memory | PATCH | `/api/v1/memory/profile`
 health | GET | `/api/v1/metrics`
 orgs | POST | `/api/v1/orgs/invite`
 orgs | GET | `/api/v1/orgs/invites`
 orgs | DELETE | `/api/v1/orgs/invites/{invite\_id}`
 orgs | GET | `/api/v1/orgs/invites/{token}`
 orgs | POST | `/api/v1/orgs/invites/{token}/accept`
 orgs | GET | `/api/v1/orgs/me`
 health | GET | `/api/v1/ping`
 portal | POST | `/api/v1/portal/access`
 portal | POST | `/api/v1/portal/sign/{token}`
 portal | POST | `/api/v1/portal/upload/{token}`
 portal | GET | `/api/v1/portal/view/{token}`
 practice | GET | `/api/v1/practice/court-
 day/calendar.ics`
 practice | POST | `/api/v1/practice/court-day/import`
 practice | GET | `/api/v1/practice/court-day/mission-
 control`
 practice | POST | `/api/v1/practice/court-day/parse`
 practice | POST | `/api/v1/practice/court-day/parse-
 file`
 practice | GET | `/api/v1/practice/court-
 day/prep/{matter_id}`
 practice | GET | `/api/v1/practice/court-
 day/prep/{matter_id}/pdf`
 practice | GET | `/api/v1/practice/court-day/today`
 practice | POST | `/api/v1/practice/court-sync`

practice | GET | `/api/v1/practice/court-sync/history`

practice | GET | `/api/v1/practice/court-sync/settings`

practice | PUT | `/api/v1/practice/court-sync/settings`

practice | GET | `/api/v1/practice/court-sync/status`

practice | GET |
`/api/v1/practice/ecourts/case/{cnr}`

practice | POST |
`/api/v1/practice/ecourts/case/{cnr}/sync`

practice | GET |
`/api/v1/practice/ecourts/causelist/available-dates`

practice | GET | `/api/v1/practice/ecourts/court-structure/states`

practice | GET | `/api/v1/practice/ecourts/court-structure/states/{state}/districts`

practice | GET | `/api/v1/practice/ecourts/search`

practice | GET | `/api/v1/practice/evidence-desk`

practice | GET | `/api/v1/practice/evidence-desk/export`

practice | POST | `/api/v1/practice/evidence-desk/scan`

practice | POST | `/api/v1/practice/evidence-desk/scan-all`

practice | POST | `/api/v1/practice/limitation/add-to-matter`

practice | POST |
`/api/v1/practice/limitation/calculate`

practice | GET |
`/api/v1/practice/limitation/presets`

practice | POST | `/api/v1/practice/litigation/ai`

practice | GET |
`/api/v1/practice/litigation/analytics`

practice | GET |
`/api/v1/practice/litigation/calendar`

practice | GET |
`/api/v1/practice/litigation/dashboard`

practice | GET |
`/api/v1/practice/litigation/diagnostics`

practice | GET |
`/api/v1/practice/litigation/hearings`

practice | POST |
`/api/v1/practice/litigation/hearings`

practice | PATCH |
`/api/v1/practice/litigation/hearings/{hearing\_id}`

practice | GET |
`/api/v1/practice/litigation/limitation/deadlines`

practice | GET |
`/api/v1/practice/litigation/notifications`

practice | GET | `/api/v1/practice/litigation/orders`

practice | POST |
`/api/v1/practice/litigation/orders`

practice | DELETE |
 `/api/v1/practice/litigation/orders/{order\_id}`
 practice | PATCH |
 `/api/v1/practice/litigation/orders/{order\_id}`
 practice | GET | `/api/v1/practice/litigation/tasks`
 practice | POST | `/api/v1/practice/litigation/tasks`
 practice | DELETE |
 `/api/v1/practice/litigation/tasks/{task\_id}`
 practice | PATCH |
 `/api/v1/practice/litigation/tasks/{task\_id}`
 practice | GET | `/api/v1/practice/litigation/war-
 room/{matter_id}`
 practice | GET |
 `/api/v1/practice/litigation/watchlist-dashboard`
 practice | GET | `/api/v1/practice/overview`
 practice | POST | `/api/v1/practice/public-intake`
 research_log | POST | `/api/v1/research/expand`
 research_log | POST | `/api/v1/research/feedback`
 research_log | GET | `/api/v1/research/history`
 research_log | POST | `/api/v1/research/log`
 saas_metrics | GET | `/api/v1/saas-metrics/kpi`
 scim | GET | `/api/v1/scim/v2/Users`
 scim | POST | `/api/v1/scim/v2/Users`
 sessions | GET | `/api/v1/sessions/by-
 id/{session_id}`
 sessions | DELETE | `/api/v1/sessions/history`
 sessions | GET | `/api/v1/sessions/history`
 sessions | DELETE |
 `/api/v1/sessions/threads/{thread\_id}`
 sessions | GET |
 `/api/v1/sessions/threads/{thread\_id}`
 sessions | DELETE |
 `/api/v1/sessions/threads/{thread\_id}/attachment`
 sessions | GET |
 `/api/v1/sessions/threads/{thread\_id}/attachment`
 sessions | POST |
 `/api/v1/sessions/threads/{thread\_id}/attachment`
 speech | POST | `/api/v1/speech/polish`
 speech | GET | `/api/v1/speech/status`
 speech | POST | `/api/v1/speech/transcribe`
 sso | POST | `/api/v1/sso/callback`
 sso | GET | `/api/v1/sso/login`
 sso | GET | `/api/v1/sso/status`
 subscriptions | GET |
 `/api/v1/subscriptions/payments`
 subscriptions | GET | `/api/v1/subscriptions/plans`
 subscriptions | GET | `/api/v1/subscriptions/portal`
 subscriptions | GET | `/api/v1/subscriptions/status`
 subscriptions | POST |
 `/api/v1/subscriptions/subscribe`
 templates | GET | `/api/v1/templates`

templates | POST | `/api/v1/templates`
templates | GET | `/api/v1/templates/{template\_id}`
templates | POST |
`/api/v1/templates/{template\_id}/generate`
trust | GET | `/api/v1/trust/account`
trust | GET | `/api/v1/trust/transactions`
trust | POST | `/api/v1/trust/transactions`

Appendix W - Complete Environment Variables Catalog

Every variable from project env templates.
Set secrets on server only; never commit
real keys.

W..env.example

****Source file:**** ``.env.example`` - Root
application (`.env`)

Variable | Example value | Notes

```
`LLM_BACKEND` | `ollama` | =====  
=====
```

LA

```
`CLOUD_GEMINI_KB` | `0` | Laptop/local: keep 0. AWS  
EC2 only: CLOUD_GEMINI_KB=1 (see  
deploy/aws/.env.produ
```

```
`LM_STUDIO_URL` | `http://127.0.0.1:1234` |  
`LM_STUDIO_MODEL` | `meta-llama-3.1-8b-instruct` |  
`OLLAMA_BASE_URL` | `http://127.0.0.1:11434` |  
`OLLAMA_MODEL` | `legalease-tuned` |  
`LLM_ROUTER_ENABLED` | `1` | Task router - defaults  
to OLLAMA_MODEL (legalease-tuned) for every Ollama  
task  
`OLLAMA_MODEL_LEGAL` | `legalease-tuned` |  
`OLLAMA_MODEL_FAST` | `legalease-tuned` |  
`OLLAMA_MODEL_LEGAL_FALLBACK` | `legalease-tuned` |  
`LLM_CLASSIFY_MAX_TOKENS` | `320` | Optional Qwen  
(only if you explicitly set different FAST vs LEGAL  
models): OLLAM  
`LLM_CLASSIFY_TIMEOUT_SEC` | `8` |  
`LLM_LEGAL_TIMEOUT_SEC` | `90` |  
`LLM_INTAKE_USE_RAG` | `1` |  
`LLM_INTAKE_RAG_K` | `3` |  
`LLM_INTAKE_FULL_ANALYSIS` | `1` |  
`INTAKE_PUBLIC_ENABLED` | `0` | Public website intake  
-> assign leads to firm user/org  
`INTAKE_PUBLIC_KEY` | `` |  
`INTAKE_ORG_USER_ID` | `` |  
`OPENROUTER_API_KEY` | `` | Web intelligence fallback  
chain (after Gemini quota/errors)  
`OPENROUTER_MODEL` | `google/gemma-2-9b-it:free` |  
`DEEPSEEK_API_KEY` | `` |  
`QWEN_API_KEY` | `` |  
`DASHSCOPE_API_KEY` | `` |  
`GEMINI_API_KEY` | `` | =====  
===== G
```

``GEMINI_FREE_MODEL` | `gemini-2.5-flash` |`
``WEB_INTELLIGENCE_DEBUG` | `0` |`
``STRICT_CITATIONS` | `0` |`
``LEGACY_WEB` | `0` |`
``JURISPRUDENCE_KB_K` | `14` |`
``GEMINI_DAILY_FREE` | `15` |`
``GEMINI_DAILY_PRO` | `200` |`
``GEMINI_DAILY_LEGAL_PRO` | `1000` |`
``GEMINI_OLLAMA_TUNING` | `0` | Settings-only: Gemini coaches Ollama via feedback analysis (never used in KB cha`
``GEMINI_COACH_MODEL` | `gemini-2.5-flash` |`
``GEMINI_COACH_FEEDBACK_LIMIT` | `25` |`
``COACH_AUTO_SCHEDULE` | `1` | Scheduled auto-coaching (daily when new feedback exists)`
``COACH_AUTO_INTERVAL_DAYS` | `1` |`
``COACH_AUTO_MIN_NEW_FEEDBACK` | `1` |`
``COACH_AUTO_CHECK_INTERVAL_SEC` | `1800` |`
``COACH_AUTO_EXPORT_MODELFILE` | `1` |`
``COACH_AUTO_ENABLE_ON_FEEDBACK` | `1` |`
``IMPROVEMENT_AUTO` | `1` | Full improvement automation - feedback -> neural train -> KB re-index -> ollama cre`
``OLLAMA_AUTO_REINDEX` | `1` |`
``OLLAMA_AUTO_CREATE` | `1` |`
``OLLAMA_AUTO_USE_TUNED` | `1` |`
``OLLAMA_AUTO_EXPORT_MIN_THUMBS` | `20` |`
``OLLAMA_TUNED_MODEL_NAME` | `legalease-tuned` |`
``OLLAMA_CREATE_TIMEOUT_SEC` | `900` |`
``GEMINI_KB_SYNTHESIS` | `0` | KB: Gemini NEVER answers or modifies retrieval (Ollama + indexed PDFs only)`
``GEMINI_KB_RETRIEVAL_HINTS` | `0` |`
``GEMINI_KB_RERANK` | `0` |`
``KB_BLOCK_RUNTIME_COACH` | `1` |`
``KB_BLOCK_LEARNING_INJECT` | `1` |`
``TAVILY_API_KEY` | `` | Gemini is allowed ONLY for Settings feedback/tuning (thumbs, coach, neural expor`
``TAVILY_SEARCH_URL` | `https://api.tavily.com/search` |`
``LEGAL_ONLY_WEB` | `1` | When Gemini quota is exhausted, Web Intel falls back to Tavily -> Serp -> Google C`
``WEB_INTEL_FAST` | `1` | Practice SaaS - public intake form (website -> CRM) INTAKE_PUBLIC_ENABLED=1 INTAK`
``WEB_INTEL_USE_LLM` | `0` |`
``WEB_SKIP_TAVILY_MCP` | `1` |`
``WEB_PREFER_TAVILY_REST` | `1` |`
``WEB_SEARCH_MAX_RESULTS` | `6` |`
``TAVILY_REST_TIMEOUT` | `12` |`

``TAVILY_MCP_TIMEOUT` | `8` |`
``WEB_LLM_MAX_TOKENS_FAST` | `900` |`
``WEB_LLM_MAX_TOKENS_CASE` | `1100` |`
``OCR_ENABLED` | `1` | OCR (EasyOCR)`
``OCR_LANGUAGES` | `en` |`
``OCR_GPU` | `0` |`
``GOOGLE_API_KEY` | `` | Google Custom Search JSON API`
 (optional - legacy; new Google CSE signups often b
``GOOGLE_CSE_ID` | `` |`
``SERP_API_KEY` | `` | SerpAPI - Google results`
 fallback when Tavily fails (<https://serpapi.com>)
``SERP_TIMEOUT` | `10` |`
``SERP_ENGINE` | `google` |`
``HF_EMBEDDING_MODEL` | `BAAI/bge-small-en-v1.5` |`
 RAG/Uploads Embeddings for RAG (never use LLM for
 vectors)
``HF_EMBEDDING_FALLBACK` | `BAAI/bge-base-en-v1.5` |`
``RAG_PREFER_BASE_EMBEDDINGS` | `1` | 1 = fast MiniLM`
 at startup/KB (recommended). 0 = load fine-tuned
 weights (slower
``RAG_USE_LANGCHAIN_HF` | `0` | 0 =`
 SentenceTransformer only (recommended on Windows;
 avoids WinError 1455 from
``RAG_SCORE_THRESHOLD` | `1.6` |`
``RAG_CONFIDENCE_THRESHOLD` | `0.52` |`
``RAG_ENABLE_CROSS_ENCODER` | `0` | 0 = fast heuristic`
 reranking (recommended on CPU). 1 = cross-encoder
 (slower, sl
``RAG_RERANK_MODEL` | `cross-encoder/ms-marco-`
 MiniLM-L-6-v2` |
``RAG_RERANK_POOL_SIZE` | `12` |`
``MAX_UPLOAD_MB` | `200` |`
``PDF_MAX_PAGES` | `0` | Large PDFs: 0 = no page cap;`
 sparse OCR only on low-text pages (fast)
``OCR_MAX_PAGES` | `0` |`
``OCR_SPARSE_ONLY` | `1` |`
``OCR_WORKERS` | `4` |`
``OCR_MIN_CHARS_PER_PAGE` | `120` |`
``RAG_CHUNK_SIZE` | `1000` |`
``RAG_CHUNK_OVERLAP` | `200` |`
``RAG_MAX_CHUNK` | `1400` |`
``RAG_INDEX_EMBED_BATCH` | `128` | Chunks embedded per`
 batch during index build (64 = faster; lower if RAM
 is tight
``RAG_FAST_INDEX` | `1` |`
``INDEX_JOB_WORKERS` | `1` |`
``INDEX_JOB_USE_PROCESS` | `1` |`
``RAG_MAX_QUERY_EXPANSIONS` | `5` |`
``RAG_TOP_K_KEYWORD` | `24` |`
``FAISS_VS_CACHE_MAX` | `8` |`
``OLLAMA_KB_LOCK_MODEL` | `1` |`

```

`PDF_UPLOAD_TIMEOUT_SEC` | `900` |
`KB_PIPELINE_DEBUG` | `1` | KB pipeline
`KB_CACHE_TTL_SEC` | `300` |
`KB_CACHE_MAX_ENTRIES` | `256` |
`RATE_LIMIT_ENABLED` | `1` | API rate limiting
`RATE_LIMIT_PER_MINUTE` | `120` |
`RATE_LIMIT_CHAT_PER_MINUTE` | `40` |
`RATE_LIMIT_CHAT_EXEMPT` | `1` | Messaging exempt by
default - Firm Chat, voice notes, AI assistant (no
429)
`RATE_LIMIT_COLLAB_EXEMPT` | `1` |
`SECURITY_HEADERS_ENABLED` | `1` | =====
===== S
`FORCE_HTTPS` | `1` |
`HSTS_MAX_AGE` | `31536000` |
`DATA_ENCRYPTION_KEY` | `` | Fernet key for optional
field encryption at rest: py -c "from
backend.app.core.c
`FIREWALL_ENABLED` | `0` | Application-layer IP
allowlist (comma-separated). Off by default. If
FIREWALL_EN
`FIREWALL_ALLOWED_IPS` | `` |
`FIREWALL_TRUST_PROXY` | `1` |
`CORS_ALLOW_LOCALHOST_REGEX` | `0` | Production:
disable localhost CORS regex (strict HTTPS origins
only)
`PASSWORD_MIN_LENGTH` | `12` |
`RAG_RETRIEVAL_K` | `8` | Retrieval depth (defaults
in code: comparison k=10, summary k=12)
`RAG_FINAL_TOP_K` | `10` |
`RAG_TOP_K_DENSE` | `16` |
`RAG_MMR_LAMBDA` | `0.7` |
`LEGALEASE_DB_PATH` | `` | =====
=====
D
`REDIS_URL` | `redis://127.0.0.1:6379/0` |
`SESSION_TTL_SEC` | `86400` |
`JWT_SECRET` | `change-me-in-production` |
`PUBLIC_APP_URL` | `http://localhost:3000` | Client
portal / e-sign / public app URL for mock sign links
`ESIGN_PROVIDER` | `mock` |
`LLM_FINETUNE_ENABLED` | `1` | In-app LLM LoRA
SFT/DPO + inference-time RLHF/RLAIF
`LLM_FINETUNE_AUTO` | `1` |
`LLM_FINETUNE_BASE_MODEL` | `google/gemma-2-2b-it` |
`LLM_FINETUNE_MIN_SFT` | `5` |
`LLM_FINETUNE_MIN_DPO` | `2` |
`LLM_USE_TRAINED_ADAPTER` | `1` |
`INFERENCE_REWARD_ENABLED` | `1` |
`INFERENCE_REWARD_RERANK` | `1` |

```

```

`CHAT_COACH_RUNTIME` | `1` |
`CHAT_COACH_POSITIVE_EVERY` | `3` |
`GPU_PROFILE` | `balanced` | GPU profile for 6GB VRAM
laptops (RTX 4050): balanced / max_stt / max_chat -
see
`STT_ENABLED` | `1` | Speech-to-text (faster-whisper;
lazy-loaded on first mic press - requires ffmpeg)
`STT_ENGINE` | `faster_whisper` |
`STT_MODEL` | `small` |
`STT_DEVICE` | `cuda` |
`STT_COMPUTE_TYPE` | `float16` |
`STT_PRELOAD` | `0` |
`STT_MAX_SECONDS` | `90` |
`STT_MAX_UPLOAD_MB` | `12` |
`STT_FALLBACK_BROWSER` | `1` |
`STT_POLISH_DEFAULT` | `0` |
`KB_LLM_TEMPERATURE` | `0.23` | KB strict grounding
(Ollama paraphrase only - documents are source of
truth)
`RAG_MIN_ACCEPT_SCORE` | `0.50` |
`KB_INDEX_MIN_VECTORS_WARN` | `20` |
`KB_LLM_TOP_P` | `0.1` |
`SAAS_PRODUCTION` | `0` | =====
===== P
`SAAS_PRODUCTION_STRICT` | `1` | Set
SAAS_PRODUCTION=1 on public deploy - enforces Stripe,
blocks mock billing, v
`ALLOW MOCK BILLING` | `1` |
`SAAS_AUTO_POSTGRES_LEGACY` | `1` | With
DATABASE_URL=postgresql://..., auto-read auth/chat
from Postgres in product
`SAAS_USE_POSTGRES_LEGACY` | `0` |
`ML_USE_QUEUE` | `1` | ML jobs off API thread
(requires Redis + py scripts/ml_worker.py or docker
ml-wo
`REDIS_URL` | `redis://127.0.0.1:6379/0` |
`PLAN_DOC_LIMIT_FREE` | `2` |
`PLAN_DOC_LIMIT_PRO` | `500` |
`PLAN_DOC_LIMIT_LEGAL_PRO` | `5000` |
`JWT_SECRET` | `change_this_jwt_secret_min_32_chars`
|
`LEGALEASE_API_SECRET` | `` |
`REDIS_URL` | `redis://127.0.0.1:6379/0` |
`PUBLIC_APP_URL` | `http://localhost:3000` |
`CORS_ORIGINS` |
`http://localhost:3000,http://127.0.0....` |
`STRIPE_SECRET_KEY` | `` | Stripe (subscription
payments - /settings/subscription)
`STRIPE_WEBHOOK_SECRET` | `` |
`STRIPE_PRICE_PRO` | `` |
`STRIPE_PRICE_LEGAL_PRO` | `` |

```

```

`STRIPE_PRICE_PRO_INR` | `999` |
`STRIPE_PRICE_LEGAL_PRO_INR` | `4999` |
`STRIPE_SUCCESS_PATH` |
`/settings/subscription?checkout=success` |
`STRIPE_CANCEL_PATH` |
`/settings/subscription?checkout=cancel` |
`MATTER_STRICT_SCOPE_ENFORCEMENT` | `1` | Matter /
org hardening
`MATTER_STRICT_ROLE_WRITE` | `1` |
`LEARNING_SCOPE_PROMOTION_ENABLED` | `0` |
`ORG_SEATS_FREE` | `1` |
`ORG_SEATS_PRO` | `3` |
`ORG_SEATS_LEGAL_PRO` | `10` |
`EMAIL_PROVIDER` | `console` | =====
=====
=== P
`EMAIL_FROM` | `noreply@your-domain.com` |
`EMAIL_FROM_NAME` | `LegalEase` |
`SUPERADMIN_USERNAMES` | `admin` | =====
=====
===== P

```

W..env.local.example

****Source file:**** ``.env.local.example`` -
Laptop overrides (`.env.local`)

Variable | Example value | Notes

```

`LEGALEEASE_LOCAL_DEV` | `1` |
`SAAS_PRODUCTION` | `0` |
`SAAS_PRODUCTION_STRICT` | `0` |
`SAAS_USE_POSTGRES_LEGACY` | `0` |
`SAAS_ALL_FEATURES_FREE` | `1` |
`SAAS_ALLOW_FREE_HYBRID` | `1` |
`ALLOW MOCK BILLING` | `1` |
`DATABASE_URL` | `` |
`LLM_BACKEND` | `ollama` |
`CLOUD_GEMINI_KB` | `0` |
`OLLAMA_AUTO_START` | `1` |
`CORS_ORIGINS` |
`http://localhost:3000,http://127.0.0.1:8000` |
`CORS_ALLOW_LOCALHOST_REGEX` | `1` |
`PUBLIC_APP_URL` | `http://localhost:3000` |
`NEXT_PUBLIC_API_URL` | `http://127.0.0.1:8000` |
`NEXT_PUBLIC_APP_URL` | `http://localhost:3000` |

```

W..env.docker.example

****Source file:**** ``.env.docker.example`` -
Docker Compose local production

Variable | Example value | Notes

```
`POSTGRES_USER` | `legalease` | --- Secrets (change
in production) ---
`POSTGRES_PASSWORD` | `change_this_strong_password` |
`POSTGRES_DB` | `legalease` |
`JWT_SECRET` | `change_this_jwt_secret_min_32_chars`
|
`LEGALEASE_API_SECRET` |
`change_this_api_secret_min_32_chars` |
`DATABASE_URL` |
`postgresql://legalease:change_this_st...` |
PostgreSQL - all app tables when
SAAS_USE_POSTGRES_LEGACY=1 (Docker default)
`SAAS_USE_POSTGRES_LEGACY` | `1` |
`SAAS_AUTO_POSTGRES_LEGACY` | `1` |
`SAAS_PRODUCTION` | `1` |
`LEGALEASE_DB_PATH` | `/data/legalease.db` | Optional
SQLite fallback path (unused when legacy Postgres is
on)
`REDIS_URL` | `redis://redis:6379/0` | Redis -
sessions + ML/e-discovery job queues
`SESSION_TTL_SEC` | `86400` |
`ML_USE_QUEUE` | `1` |
`NEXT_PUBLIC_API_URL` | `https://your-domain.com/api`
| Frontend must call API through nginx (/api prefix)
`CORS_ORIGINS` | `https://your-domain.com` |
`PUBLIC_APP_URL` | `https://your-domain.com` |
`LLM_BACKEND` | `lmstudio` | LLM - LM Studio on host
or Ollama sidecar
`LM_STUDIO_URL` | `http://host.docker.internal:1234`
|
`LM_STUDIO_MODEL` | `meta-llama-3.1-8b-instruct` |
`GEMINI_API_KEY` | `` | Gemini (Open Law / Hybrid)
`GEMINI_DAILY_FREE` | `15` |
`GEMINI_DAILY_PRO` | `200` |
`GEMINI_DAILY_LEGAL_PRO` | `1000` |
`EMAIL_PROVIDER` | `console` | Email (console / smtp
/ sendgrid)
`EMAIL_FROM` | `noreply@your-domain.com` |
`SMTP_HOST` | `` |
`SMTP_PORT` | `587` |
`SMTP_USER` | `` |
`SMTP_PASSWORD` | `` |
`SENDGRID_API_KEY` | `` |
`STRIPE_SECRET_KEY` | `` | Stripe subscriptions
```



```

`STRIPE_WEBHOOK_SECRET` | `` |
`STRIPE_PRICE_PRO` | `` |
`STRIPE_PRICE_LEGAL_PRO` | `` |
`SUPERADMIN_USERNAMES` | `admin` | Admin + monitoring
`SENTRY_DSN` | `` |
`SENTRY_ENVIRONMENT` | `production` |
`SENTRY_TRACES_SAMPLE_RATE` | `0.1` |
`RAG_ENABLE_CROSS_ENCODER` | `0` | Performance
`KB_PIPELINE_DEBUG` | `0` |
`RATE_LIMIT_ENABLED` | `1` |
`TAVILY_API_KEY` | `` |

```

W.deploy/aws/.env.production.example

****Source file:****

`deploy/aws/.env.production.example` - EC2
production template

Variable | Example value | Notes

```

`POSTGRES_USER` | `legalease` |
`POSTGRES_PASSWORD` | `(set on server - see section
7.4.5)` |
`POSTGRES_DB` | `legalease` |
`JWT_SECRET` | `(set on server - rotate_secrets.ps1)`
|
`LEGALEASE_API_SECRET` | `(set on server -
rotate_secrets.ps1)` |
`DATA_ENCRYPTION_KEY` | `(set on server -
rotate_secrets.ps1)` |
`DATABASE_URL` |
`postgresql://legalease:POSTGRES_PASSW...` |
`SAAS_USE_POSTGRES_LEGACY` | `1` |
`SAAS_AUTO_POSTGRES_LEGACY` | `1` |
`SAAS_PRODUCTION` | `1` |
`SAAS_PRODUCTION_STRICT` | `1` |
`ALLOW MOCK BILLING` | `0` |
`LEGALEASE_DB_PATH` | `/data/legalease.db` |
`REDIS_URL` | `redis://redis:6379/0` |
`SESSION_TTL_SEC` | `86400` |
`ML_USE_QUEUE` | `1` |
`PUBLIC_APP_URL` | `https://legalease.duckdns.org` |
`CORS_ORIGINS` | `https://legalease.duckdns.org` |
`CORS_ALLOW_LOCALHOST_REGEX` | `0` |
`NEXT_PUBLIC_API_URL` |
`https://legalease.duckdns.org/api` |
`LLM_BACKEND` | `gemini` |
`GEMINI_API_KEY` | `(set on server -
rotate_secrets.ps1)` |
`GEMINI_KB_SYNTHESIS` | `0` |

```

```
`CLOUD_GEMINI_KB` | `1` |  
`LEGALEEASE_HF_CACHE` | `/data/hf_cache` |  
`HF_HOME` | `/data/hf_cache` |  
`LOW_RESOURCE_MODE` | `1` |  
`RAG_ENABLE_CROSS_ENCODER` | `0` |  
`EMAIL_PROVIDER` | `brevo` |  
`BREVO_API_KEY` | `` |  
`STRIPE_SECRET_KEY` | `` |  
`STRIPE_WEBHOOK_SECRET` | `` |  
`STRIPE_PRICE_PRO` | `` |  
`STRIPE_PRICE_LEGAL_PRO` | `` |  
`FORCE_HTTPS` | `1` |  
`SECURITY_HEADERS_ENABLED` | `1` |  
`RATE_LIMIT_ENABLED` | `1` |  
`FIREWALL_ENABLED` | `0` |  
`FIREWALL_ALLOWED_IPS` | `` |  
`FIREWALL_TRUST_PROXY` | `1` |  
`SUPERADMIN_USERNAMES` | `admin` |
```

*Complete documentation built 2026-06-05
17:35 UTC. Production URL:
<https://legalease.duckdns.org>*